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The Luminosity Function of Ultrafaint Trans-Neptunian Objects Detected by JWST

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Abstract

We present a definitive discovery of 27 trans-Neptunian objects (TNOs) using the Near-Infrared Camera aboard the James Webb Space Telescope (JWST). By employing a shift-and-stack technique and a machine learning network geared specifically to identifying false-positive detections in JWST images produced through the shift-and-stack process, we achieved a 40% detection threshold of $m_{\rm F150W2} = 28.8$ mag (corresponding to $m_r \sim 29.8$ mag) across a sky area of 0.05 deg^2. This marks the deepest solar system survey to date, reaching magnitudes that allow us to explore never-before-seen regions of the TNO size distribution. Our faintest detection has $m_{\rm F150W2} = 29.3$ mag and a diameter of ∼10 km (assuming 15% albedo). Within our sample, we find that both the cold and hot TNO subpopulations exhibit a power-law slope. The distribution of apparent magnitudes of our nominal sample (detections at all epochs) is well fit by a single power law $dN/dm \propto 10^{\alpha m}$ with $\alpha = 0.29^{+0.08}_{-0.07}$. The dynamically hot and cold subsamples in our discovery set are consistent with the same power law, suggesting that the planetesimal formation process yields similar slopes despite the differing disk conditions at the presumed ∼25 and ∼45 au formation regions of the two populations.

Unified Astronomy Thesaurus concepts: Trans-Neptunian objects (1705); Kuiper belt (893); Small Solar System bodies (1469)

1. Introduction

The discovery three decades ago (D. C. Jewitt & J. X. Luu 1993) of the second trans-Neptunian object (TNO[9])—after Pluto—opened an entirely new window into the origin and evolutionary history of our solar system. These distant, icy planetesimals beyond the orbit of Neptune have undergone minimal change over the last 4 billion years and therefore serve as natural laboratories for studying early solar system processes and testing models of planet formation. Dynamically, TNOs can be broadly divided into two categories: the quiescent "cold" classicals (CCs), which reside on low-inclination, nearly circular orbits, and the excited "hot" population (HTNOs), which instead occupy more eccentric, more highly inclined trajectories.

The size distribution (SD) of TNOs, in particular the CCs, which can be inferred from their luminosity function (LF), provides key constraints on the conditions of planetesimal formation as well as the evolutionary processes that shaped them (B. Gladman et al. 2001; G. M. Bernstein et al. 2004; W. C. Fraser & J. Kavelaars 2008; W. C. Fraser et al. 2008, 2010; W. C. Fraser et al. 2014; J. Kavelaars et al. 2021; J.-M. Petit et al. 2023; K. J. Napier et al. 2024). Comparing the observed SD with those predicted by planet formation models helps to constrain the proposed physical processes and underlying initial conditions that shaped the current solar system. However, the relative faintness and the typical 35–50 au distances of TNOs limit ground-based surveys to about $m_r \sim 27$ mag ($D \sim 50$ km) (W. C. Fraser & J. J. Kavelaars 2009; C. I. Fuentes et al. 2009). The most sensitive search to date was carried out with the Hubble Space Telescope (HST) and reached $m_r \sim 28$ mag ($D \sim 30$ km) (G. M. Bernstein et al. 2004). This HST program covered 0.02 deg^2 and detected only three objects, far fewer than the ∼85 objects expected from extrapolations of the bright end of the LF. More recently, an HST search for potential extended New Horizons mission targets identified only two objects of diameters ∼40 km (M. W. Buie et al. 2024), consistent with extrapolations from W. C. Fraser et al. (2014) for that survey area of 0.3 deg^2. With so few detections, our knowledge of the TNO SD at small sizes remains poorly constrained, and consequently, our understanding of planetesimal formation and growth in the outer solar system remains limited.

The scenario of a streaming instability (SI) followed by pebble cloud collapse (PCC) has recently emerged as the leading model for planetesimal formation. In this model, aerodynamic interactions between gas and dust concentrate pebble-sized bodies into dense, self-gravitating clouds, which then collapse rapidly to form planetesimals on the order of ∼100 km in size (A. N. Youdin & J. Goodman 2005; A. Johansen & M. Lambrechts 2017; U. Schäfer et al. 2017; D. Nesvorný et al. 2021). Several lines of evidence support this model, including the exceptionally low

[9] Often used interchangeably with Kuiper Belt object, or KBO.

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bulk densities observed among medium- to large-sized TNOs ($\lesssim$1000 km; W. Grundy et al. 2019; X. Mao et al. 2021; J. T. Keane et al. 2022), the high binary fraction across most TNO populations (K. S. Noll et al. 2020), the compositional similarities frequently found between components of binary systems (S. D. Benecchi et al. 2009; M. Marsset et al. 2020), and dynamical studies showing that the SI+PCC mechanism naturally produces a binary orbital distribution consistent with observations (W. Grundy et al. 2019; D. Nesvorný et al. 2019). Moreover, most recently, J. Kavelaars et al. (2021), J.-M. Petit et al. (2023), and K. J. Napier et al. (2024) found that the SD of TNOs is consistent with an exponentially tapered power law predicted by the SI +PCC mechanism (C. P. Abod et al. 2019; R. Li et al. 2019; U. Schäfer et al. 2017; J. B. Simon et al. 2016). However, J. Kavelaars et al. (2021) and J.-M. Petit et al. (2023) only reached magnitudes as faint as $H_r \sim 8.3$ ($\sim$75 km), while K. J. Napier et al. (2024) extended to $H_r \sim 10.5$ ($\sim$27 km), leaving the faint-end consistency of the SI+PCC model still uncertain.

We conducted a TNO-dedicated search with the Near-Infrared Camera (NIRCam) aboard the James Webb Space Telescope (JWST) covering 0.05 deg^2 of the sky and achieved a limiting magnitude of $m_{\rm F150W2} = 28.8$ mag to explore the smaller end of the TNO SD. In this paper, we begin by presenting the observational design in Section 2, detailing the strategies implemented to maximize our detections. Section 3 describes the data reduction and processing pipeline, which includes generation and injection of fake sources, image alignment, and subtraction of the sidereal sky. Our methodology for identifying TNO candidates and fitting their orbits is presented in Sections 4 and 5, where we explain the tools and criteria used to confidently identify TNOs across multiple epochs. Section 5 summarizes our results, detailing the linking of candidates across three and two epochs, their derived orbital elements and magnitudes, and additional single-epoch detections. In Sections 7, 8, and 9, we turn to the broader implications of our detections. Section 7 presents our analysis of the luminosity function, identifying the model that best describes our TNO detections. In Section 8, we present the cumulative absolute magnitude H distribution of the CC population and compare our results with predictions from SI +PCC models. Section 9 then provides an updated estimate of the total mass of the Kuiper Belt. Finally, in Section 10, we place our findings in the context of planetesimal formation and solar system evolution models, discussing how our results inform our understanding of the formation and evolution of the cold and hot classical populations, as well as the Kuiper Belt as a whole.

2. Observing Strategy

The observing strategy balances several interrelated factors: the area to be surveyed, on-sky location of the survey, observing cadence, integration time, exposure time per epoch, number/timing of epochs, and filter choice. In combination, these determined the limiting magnitude (the proxy for minimum TNO size), expected number of detections, and accuracy of the orbit determinations.

The NIRCam instrument (M. J. Rieke et al. 2023), with its fairly large 8.9 arcmin^2 instantaneous field of view (FOV) and exceptional sensitivity, was the natural choice for the survey. NIRCam has two identical halves (modules), each consisting of a short-wavelength (SW; 0.6–2.35 μm) and a long-wavelength (LW; 2.4–5.0 μm) channel that simultaneously image the same FOV. The FOVs of the two modules are separated by about 40″. The SW channel in each module is populated with four 2048^2 HgCdTe detectors with 4″ gaps between them, while the LW channel in each module consists of one such detector. The pixel scales in the SW and LW channels are 31 and 64 mas, respectively. The rectangle bounding the FOVs of the two modules is 4.′95 × 2.′14, with the long axis perpendicular to the ecliptic due to roll restrictions imposed by the need to keep the telescope and instruments behind the Sun shield.

To maximize sensitivity, we selected the F150W2 and F322W2 filters in the SW and LW channels. The bandwidth ($\delta\lambda/\lambda$) of these filters is 0.82 and 0.42, respectively, are the widest and hence most sensitive available in each channel. However, each provides nearly diffraction-limited FWHM point-spread functions (PSFs) in each channel. For an initial reconnaissance of the smallest objects ever detected in the trans-Neptunian region, they are optimal.

The observations used sidereal tracking of a field centered at (R.A., decl.) of (13:57:33, −10:51:55), which is on the solar system's invariable plane (1.°1 north of the ecliptic plane at that ecliptic longitude), and in a region of very low stellar density. The number of TNOs to be discovered depends on the area covered by the survey, the true magnitude (or size) distribution of the TNOs (which is not known!), and the detection limit. Our modeling led us to select a mosaic of 20 tiles of the NIRCam FOV as the optimal trade between area and depth for a nominal prediction of the magnitude distribution. Figure 1 illustrates the 20-tile mosaic, and the coverage of the individual detectors for one tile. The full mosaic was observed at each of three epochs, with about 5 days between epochs in order to constrain the orbital parameters, as described later. Observations were taken from 2023 January 24 through 2023 February 4.

Because of the sidereal tracking, the TNOs we sought to discover were subject to trailing in the images. The apparent rate of motion of TNOs in the classical Kuiper Belt at 45 au is $\leqslant$0.12 mas s^{-1} (0.″4 hr) when viewed from JWST within ±5 deg of 98 deg solar elongation (appropriate for our observations) and is about 3× lower as they transition between apparent prograde and retrograde motion. A TNO moving at 0.1 mas s^{-1} crosses a 31 mas SW pixel in 310 sec, and moves by the 45 mas FWHM of the F150W2 PSF in 450 s. To avoid significant image trailing, we used 204 s integrations, resulting in trailing of $\leqslant$0.7 SW pixels. Motion of a bright TNO's flux into or out of a pixel during an integration produces nonlinear ramps of counts versus time, which can be incorrectly flagged as cosmic-ray (CR) hits during science pipeline processing. Our choice of integration time is short enough that only TNOs both brighter and faster than our regime of interest would be flagged as CR hits, allowing us to run the pipeline with CR correction enabled.

We used exposures with three integrations, providing an exposure time of 612 s (reset overheads mean each such exposure takes 639 s). Two exposures dithered (by 3″) were taken at each epoch. Including the time required to execute the dither, the time between the midtimes of the first integration of the first exposure and the third integration of the second exposure is 1156 s, sufficient for a typical TNO's apparent motion to be $\sim$60 mas, i.e., two SW pixels or 1.3 × the PSF FWHM. This is easily detectable and yields a useful measure of the rate of motion of each TNO. One "epoch" consists of a

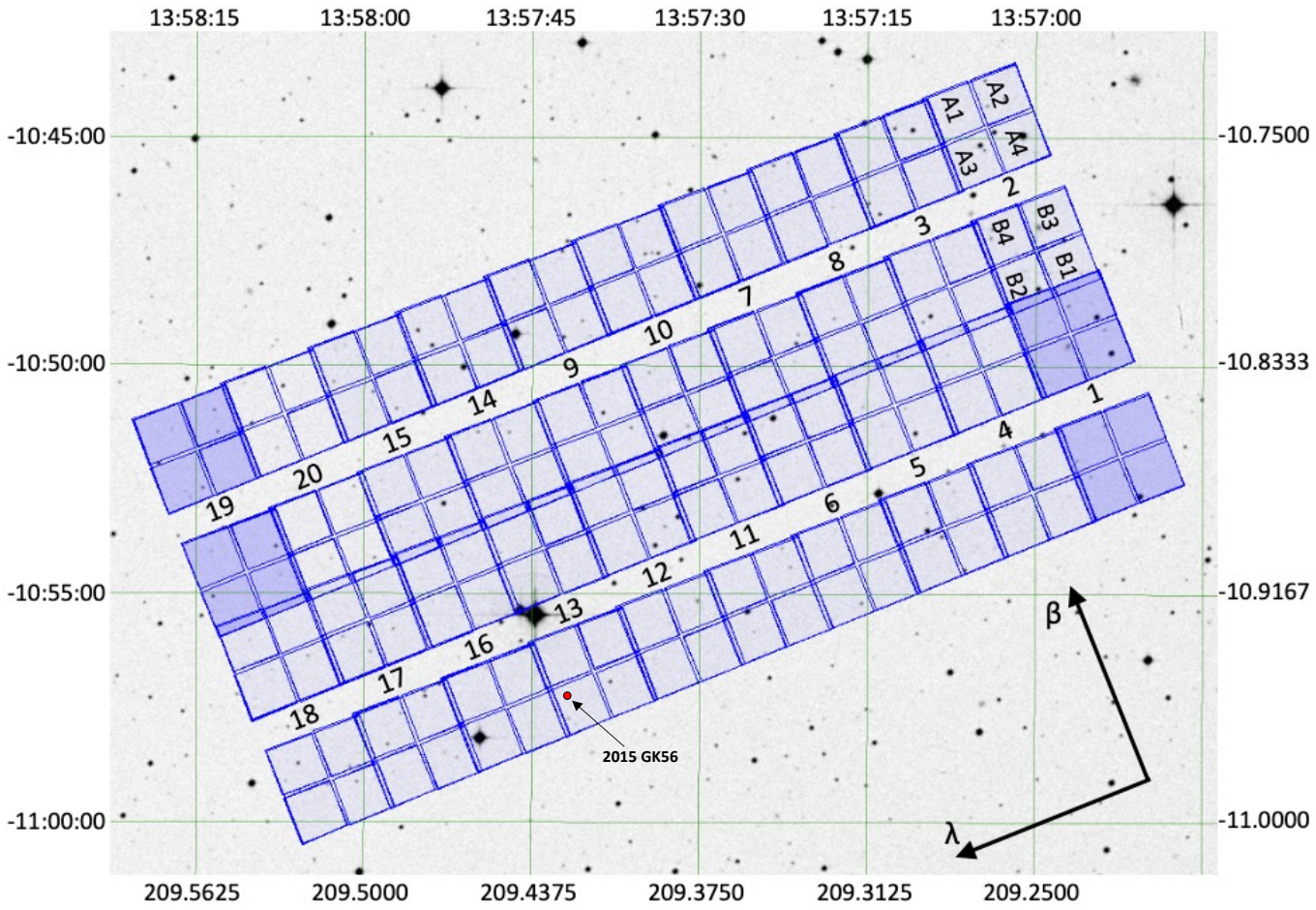


Figure 1. The mosaic used for the observations consisted of 20 tiles, each tile consisting of the footprints of the eight NIRCam SW detectors (and the two coaligned LW detectors). Darker blue shading shows the footprints of single tiles at the left and right ends of the mosaic. The relative positions of the detectors in the A and B modules are shown for the upper-right tile of the mosaic, and highlight the 40″ gap between the two modules. Numbers from 1 to 20 in the mosaic gaps label the tile ordering, equivalent to the JWST visit numbers. The sequence of tiles was repeated in three epochs separated by ≈5 days. The red circle in tile 13 is the location of a known TNO, 2015 GK56, which was purposefully included in the survey footprint to evaluate recovery performance. The coordinate labels are in the ICRS equatorial system, with the orientation of the ecliptic coordinate system shown by the vectors at lower right. The background image is from the Digital Sky Survey.

two-exposure visit to each of the 20 tiles. At a given epoch, all six integrations were acquired consecutively for a particular mosaic tile, but observations of adjacent tiles were not always contiguous in time (see Figure 1).

Taking the visible to near-infrared spectrum of Arrokoth from the New Horizons spacecraft (W. Grundy et al. 2020) as our template for the reflectance of other small CC TNOs (see Figure 2), we used the JWST Exposure Time Calculator (ETC) and determined that the ultrawide (1.0–2.35 μm) F150W2 filter gave significantly better sensitivity than any other choice in the SW channel. At the time, the spectra of TNOs (other than Pluto) at $\lambda \gtrsim 2.2$ were unknown, so we chose the very wide (2.4–4.0 μm) F322W2 filter in the LW channel.

To estimate the limiting magnitude of our search, we used the solar-convolved spectrum of Arrokoth and the JWST ETC. The expected V-band (Vega) absolute magnitude of a 10 km diameter object with a V-band geometric albedo of 0.15 is 12.7. At a distance of 45 au from JWST and from the Sun, the apparent magnitude of such a TNO is 29.4 mag. In the ETC, we normalized the spectrum described above to 29.4 mag at V, and defined the exposure parameters (also discussed above) using 10 groups of the BRIGHT1 exposure pattern, three integrations per exposure, and two (dithered) exposures. The total effective exposure time per observation is then 1267 s, and the predicted signal-to-noise ratio (S/N) was 11 in the F150W2 filter. As of ETC version 4.0, released in 2024 July, that predicted S/N is somewhat lower at 8.4. The expected S/N in a single integration is 2.4 × lower (4.4 in the original predictions, 3.5 using ETC 4.0).

The mosaic observation was repeated at three different epochs to allow us to measure the objects' intrinsic velocities as well as their reflex acceleration caused by JWST's orbital motion. We wanted the total range of apparent motion for TNOs as close as 30 au to be well below the ≈20′ extent of the observed field in the ecliptic direction, so that only a small fraction of detected TNOs will fall off the field during observations. The apparent ecliptic-plane position of a body at 30 au moves roughly $0.'17\ (\Delta t/1\,\mathrm{day})^2$ from the turnaround point, where Δt is the time from turnaround. To keep this motion to ≈1′ or less requires $\Delta t < 7$ days. Based on these considerations, we spaced three mosaic observations by 5 ± 1 days, and timed them to occur near quadrature, at $\theta_{\mathrm{Sun}} \simeq [95°, 100°, 105°]$ (±1°). The three sightings we expected for each object discovered would then allow us to determine distance to ≪1 au and inclination i to ≪1°, and set useful limits on a and e for brighter TNOs. This predicted accuracy is borne out in our results (Section 5.3). We note that given the anticipated few-mas precision of the JWST astrometry, a 10 day arc provides a similar lever on a target's ephemeris as

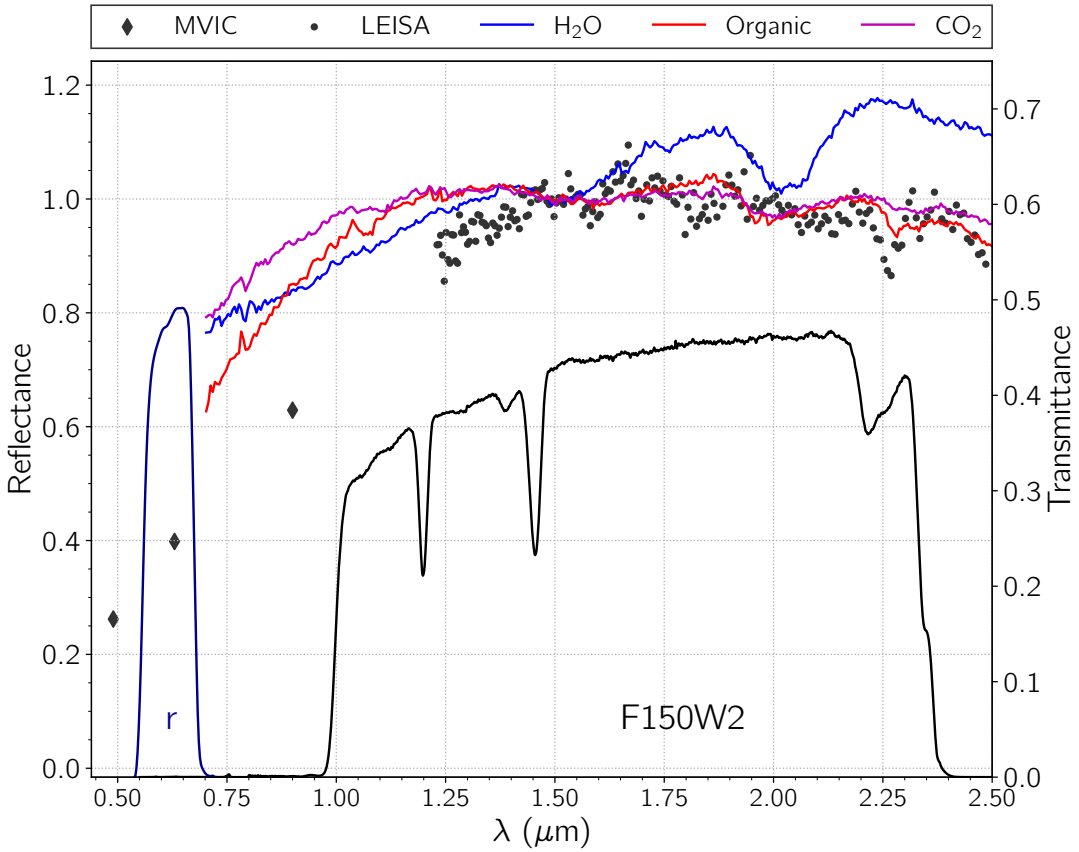


Figure 2. Median reflectance spectra of the organic, CO_2, and H_2O surface types identified in the JWST NIRSpec sample. The LEISA and MVIC I/F reflectance data for Arrokoth are shown in black points. All spectra are normalized to 1.5 μm. The transmission bandpasses of the *r*-band and F150W2 filters are shown in blue and black, respectively.

does a full one-year arc of ground-based observations in typical seeing.

3. Image Processing

3.1. Generation of Fake Sources

To determine detection areas and thresholds, we generate approximately 10,000 synthetic objects that emulate the motion, lightcurve, and color characteristics of real TNOs, distributed within a radius of $0.^{\circ}8$ centered on our observational footprint. Of these, about 900 objects fall within the mosaic area during at least one exposure, and were subsequently implanted into the raw JWST images (we discuss this in Section 3.2).

The synthetic TNOs are created using a combination of (1) realistic orbital elements informed by known objects, and (2) deliberately "unrealistic" distributions to explore hypothetical populations. For realistic objects, semimajor axis a and eccentricity e are based on data from the Minor Planet Center (MPC).[10] We start from the orbital elements of known TNOs and perturb each MPC a and e by adding a random zero-mean deviate to each—fractions of an astronomical unit for a and a few percent for e. For unrealistic objects, a is drawn from a logarithmic distribution between 30 and 300 au, and e is drawn from a uniform distribution between 0 and 1.

The absolute magnitudes H for realistic objects are drawn from the power-law distribution given by W. C. Fraser et al. (2014). In contrast, unrealistic objects are assigned H values from a uniform distribution, allowing sufficient statistics to determine the detection efficiency over a broader range of magnitudes.

We calculate the synthetic objects' apparent sky positions at one-minute intervals throughout the observation period using NASA's NAIF SPICE toolkit via the `spiceypy` Python interface (A. M. Annex et al. 2020). Only objects falling within the mosaic footprint (plus a buffer) during the observation period are retained.

After determining each object's trajectory, we model its photometric variability by assigning a sinusoidal lightcurve. Rotation periods are drawn from a Maxwellian distribution spanning 2–20 hr, peak-to-peak amplitudes are selected based on the distributions described by A. Thirouin & S. S. Sheppard (2019) for CC TNOs, and rotational phases are assigned at random. The mean magnitude of each lightcurve is set by the assigned absolute magnitude H, and the apparent magnitude m is then computed as a function of the object's motion and rotational variability. This approach produces realistic TNO brightness variations and allows us to measure our survey sensitivity not only as a function of mean apparent magnitude but also as a function of lightcurve amplitude.

3.2. Implantation of Fake Sources

The process of implantation starts by creating linearized seed images of fake sources with an image simulation software called `MiRaGe: Multi-Instrument RAmp GEnerator` (B. Hilbert et al. 2019). Linearized seed images contain only the artificial TNO sources and give readout signals up the integration ramp that are linear with time. The linearity

[10] https://minorplanetcenter.net

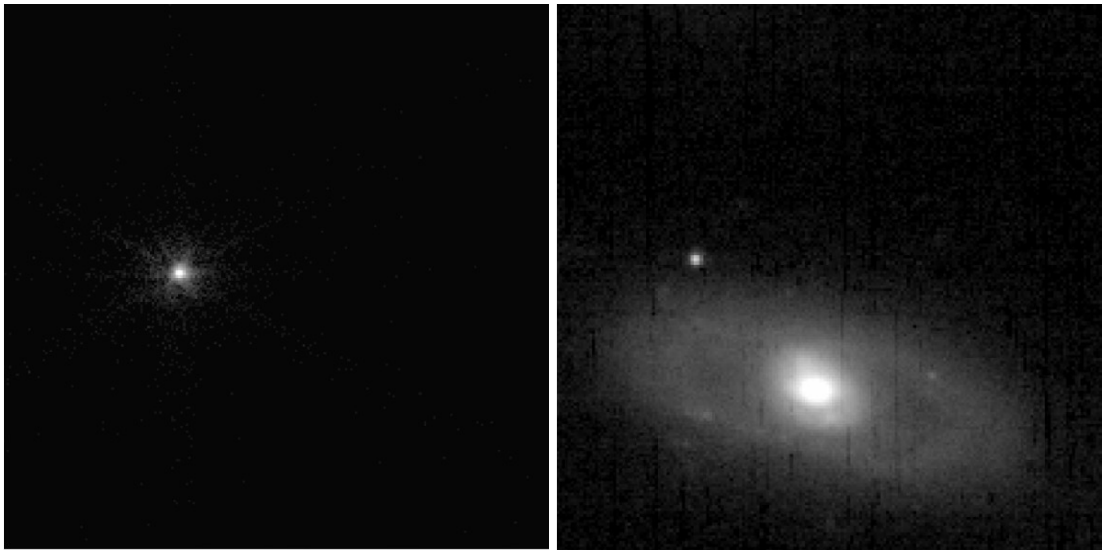


Figure 3. Left: cutout of a linearized seed image containing only the artificial TNO. Right: image output after addition of the seed image to the linearized real JWST image.

correction, which compensates for nonlinear detector response due to effects such as sensor saturation, is applied in the pipeline so that the linearized image arrays can be accurately added into the linearized real JWST images (see Figure 3). About 900 artificial sources with a magnitude range of $m_{\mathrm{F150W2}} = 21.5$–$30.5$ mag and a speed of $0\farcs03$–$8\farcs66\,\mathrm{hr}^{-1}$ were implanted across JWST integrations. A typical NIRCam SW detector image contains 6–10 artificial sources.

Once the linearized seed images are added to the linearized real JWST MULTIACCUM integration data—which consist of nondestructive readouts taken throughout an exposure—we proceed to run the slope fitting step from Stage 1 of the JWST Calibration Pipeline. This step fits a line to the signal accumulation in each pixel, converting the raw integration ramps into 2D images of count rates (i.e., the measured signal rate per pixel). The result is a `rateints` file, containing one count-rate image for each of the three integrations in the exposure.

Next, we run Stage 2 of the JWST calibration pipeline (`calwebb_image2`; H. Bushouse et al. 2022) to create a fully calibrated (`calints`) image per integration. This stage applies the flat field and performs the flux calibration, such that the output data are in units of MJy sr^{-1}. The output products at this point are "`calints`" files, with each file containing all three integrations of an exposure at a single pointing.

3.3. Image Alignment

A major task is to map all of the exposures to a self-consistent World Coordinate System (WCS). JWST data use a generalized WCS (gWCS) (N. Dencheva et al. 2024). To improve the astrometric alignment beyond the default MAST[11] products, we used a multistep process involving several calls to the `Tweakreg` step of the JWST calibration pipeline (H. Bushouse et al. 2022). `Tweakreg` improves the WCS of individual images by matching and aligning common source positions between them.

We began by aligning the longwave (LW) module images, which requires fewer detector-to-detector adjustments since each module has only one detector, whereas shortwave (SW) modules have four detectors each. The initial alignment focuses on a subset of mosaic tiles (tiles 10, 14, and 15) which show significant misalignments between epochs 2 and 3. Different guide stars were used in epoch 3 of these tiles compared to the first two epochs, so uncertainties in the absolute guide star locations propagate into differences in the WCS between the epochs. We first call `Tweakreg` individually on these tiles, in order to improve the epoch-to-epoch alignments.

After the data within each tile are well aligned, we proceed to align the tiles with respect to each other. We first perform a relative alignment of pairs of tiles in the north/south direction of the 2×10 mosaic, i.e., within a column (Figure 1). The detector overlap between tiles is largest in this direction, allowing `Tweakreg` to identify 8–16 sources in each overlapping region. Aligning with these sources typically results in a reported rms of $0\farcs007$–$0\farcs008$.

Once the tiles within each column have been aligned, we use a separate call to `Tweakreg` to align the columns to one another. With the tiles for a given column aligned, we effectively double the area of overlapping sky available when aligning adjacent columns. rms values reported by `Tweakreg` are generally $0\farcs008$–$0\farcs030$, based on the 15–20 sources in the overlap region between each column pair.

After completing the relative tile-to-tile alignment, we use `Tweakreg` to align the entire LW mosaic to the GAIA DR3 catalog in a single calculation. `Tweakreg` reports an rms of $0\farcs03$ for this step.

With the relative and absolute alignment of the LW mosaic complete, we align the SW mosaic to the LW mosaic using a single call to `Tweakreg`, aligning the SW files to a source catalog from the aligned LW mosaic. For this task, `Tweakreg` identifies 130–195 sources per tile, and calculates rms values of $0\farcs01$–$0\farcs03$ per tile, when fitting for shift, rotation, and skew.

3.4. Background Subtraction

After completing the WCS alignment, we proceed to subtract the $1/f$ noise from the images using a code[12] specifically designed for processing and analyzing JWST data.

[11] https://mast.stsci.edu

[12] https://github.com/chriswillott/jwst/blob/master/image1overf.py

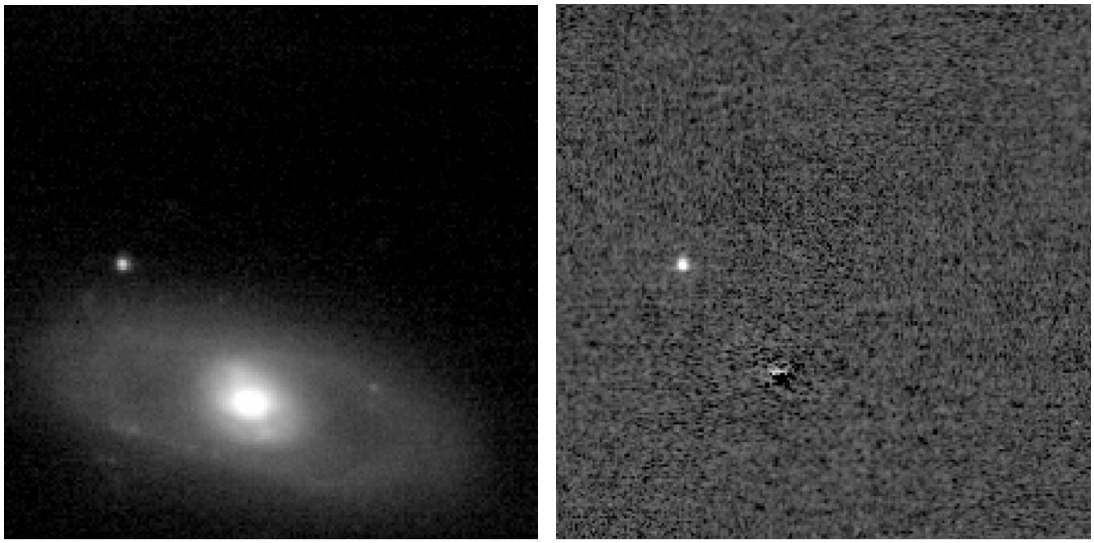


Figure 4. Left: cutout of a JWST image featuring a moving point source next to a bright galaxy. Right: same image cutout after the subtraction process.

In general, we subtract the $1/f$ noise independently from each of the four amplifiers in a given image. In some cases, however, large diffraction spikes or large extended sources result in a poor $1/f$ correction for a given amplifier. In these cases, we subtract the $1/f$ signal in all four amplifiers simultaneously.

The next step was to subtract the "background" from each of the images, where in this case, "background" refers to all sidereal sources. After subtracting the background, the only sources that should remain in the data are solar system objects and any sources with time-variable brightness. Working one tile at a time, we use the exposures from any two of the epochs to create the background image that will be subtracted from the third epoch's data. In order to create the background images, we use Stage 3 of the JWST calibration pipeline (H. Bushouse et al. 2022). This stage of the pipeline is not designed to run on `calints` files. Therefore, we split the images from the three integrations within each `calints` file into three separate `cal` files. In a single call, we run the pipeline on all of the `cal` files from two of the epochs, being sure to turn off the `Tweakreg` step since the files were previously aligned, as discussed above. By including all files for two epochs, all moving targets will not be in the same location from one file to the next, and will therefore be flagged and ignored when the images are resampled onto the output pixel grid. The result is a "background" image containing only sidereal targets; all moving targets have been removed.

We then call the Stage 3 pipeline separately on each `cal` file from the third epoch. In this case, operating on a single file each time, no sources are flagged and ignored. As part of the pipeline call, we specify that the input image be resampled onto the same pixel grid as the background image described above. The background image is then subtracted from the resampled individual image, creating an image where sidereal targets with a constant signal have been removed, and only moving targets or time-variable sources remain. We follow this recipe for all tiles, and also for each combination of epochs, resulting in a background-subtracted version of each integration for all tiles and epochs (see Figure 4).

4. Search Process

The search process in our survey is composed of five steps: (1) shift-and-stack, (2) machine learning vetting, (3) single-epoch scene modeling, (4) linking, and (5) orbit refinement and selection. In the shift-and-stack stage, we perform an initial search across the image stacks of every single epoch to identify candidate objects exhibiting sky motions consistent with a TNO. The machine learning vetting step refines the candidate list by applying a binary classification of the shift-and-stack feature output as consistent with a TNO (real) versus a false positive (spurious). During scene modeling, we verify whether the image pixel data are consistent with a model of a moving point source superimposed on a static sky background. This step also provides optimal estimates of the source's flux and position as a function of time, along with their associated uncertainties. The linking step finds all combinations of one track from each epoch that are consistent with a single TNO orbit. Finally, the orbit refinement and selection determine the best-fit orbital parameters for each linked treatment by a joint fit to all three epochs' pixel data, producing best-fitting orbital parameters and fluxes, and goodness-of-fit measures that are used to select a sample with no false positives and high completeness.

A schematic diagram outlining the sequence and interrelation of these steps is presented in Figure 5, and we discuss each of them in the following subsections.

4.1. Shift-and-Stack

The implanted, subtracted, and pixel-aligned `calints` files were searched using the digital tracking, or shift-and-stack. We followed the approach implemented in `kbmod` (P. J. Whidden et al. 2019), but rewritten to allow for some improvements. We made use of the same likelihood function as in P. J. Whidden et al. (2019) and indeed that used in G. M. Bernstein et al. (2004), but with the use of the JWST NIRCam PSF model. The two `calints` exposures were separated into their three separate integrations, preserving their variance planes. Mask planes were generated marking all bad pixels identified by the earlier processing, resulting in six separate images for the shift-and-stack.

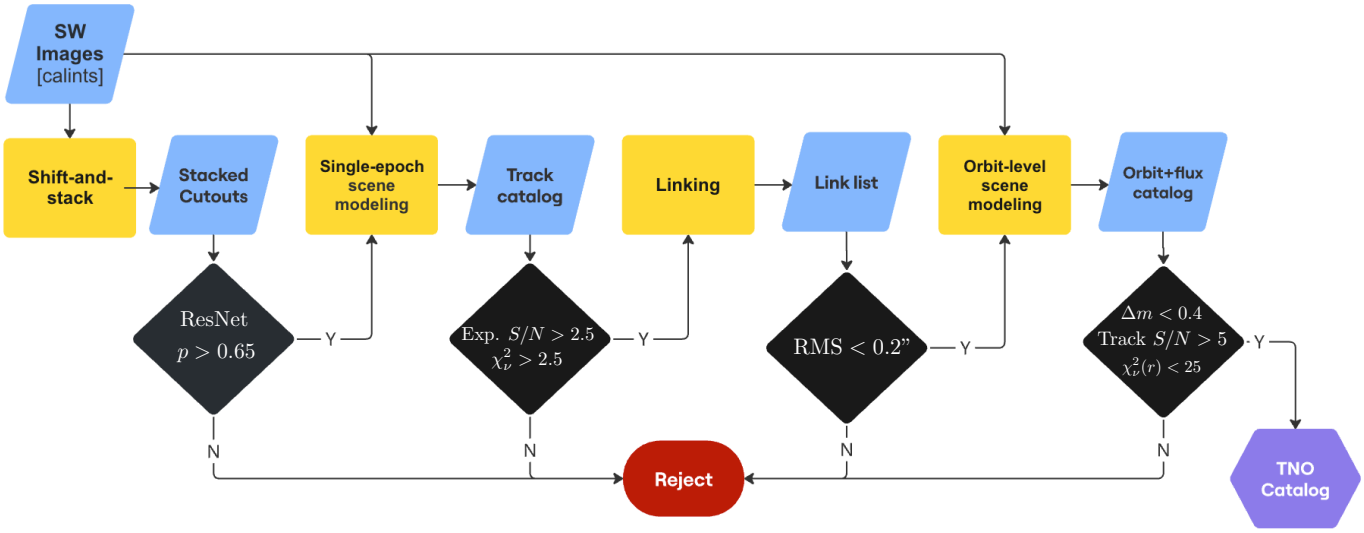


Figure 5. A chart depicting the processing workflow from calibrated and subtracted imagery to a set of detected KBOs of high purity. Black rectangles indicate stages where candidate sources (singular or linked multiple sources) are rejected. Major process stages and the resultant data products are depicted by the yellow rectangles and blue trapezoids, respectively.

A grid of linear shifts was used in the search, with the grid encompassing bound orbits outside ∼15 au, corresponding to rates of motion of $-2.6 \leqslant v_\alpha \leqslant 2.''6\,\mathrm{hr}^{-1}$ and $-1.3 \leqslant v_\beta \leqslant 1.''3\,\mathrm{hr}^{-1}$ for the first epoch, and $-1.3 \leqslant v_\alpha \leqslant 1.''3\,\mathrm{hr}^{-1}$ and $-1.3 \leqslant v_\beta \leqslant 1.''3\,\mathrm{hr}^{-1}$ for the second and third epochs. The grid spacing was $0.''16\,\mathrm{hr}^{-1}$, resulting in a maximum possible mis-shift of less than 0.5 pixels. The result of the shift–stack was a list of candidates with estimates of position and velocity vectors, and an estimate of source brightness, B_{ml}, determined from an exact solution of the likelihood model (P. J. Whidden et al. 2019).

After the initial shift–stack, any candidates with negative maximum-likelihood fluxes were rejected. Candidates then underwent numerous filters to reject bogus sources. First, a brightness filter was applied whereby at a given detection location, the candidate's maximum-likelihood estimated brightness value, B_{ml}, was varied from 0.5 to 1.5 times, sampled on a grid of nine evenly spaced values. At each brightness, the likelihood was calculated exactly. If the maximum likelihood of those nine different evaluations was either at the minimum or maximum of the grid, then the source was deemed to not behave as expected for a real moving source and was rejected. Real moving sources show behavior like assumed in the likelihood model, and so have valid brightness estimates F_{ml} and show a peak in likelihood near their true fluxes. Junk sources tend not to, and so can be removed by this filter. Increasing the density of the samples beyond nine evaluations did not eliminate any additional candidates.

The remaining candidates underwent a so-called predictive filter which was designed to take advantage of the fact that the majority of good detections (real or implanted) were detected at more than one rate. Given a source detected at a grid rate that most closely matches its true rate of motion, the median position of that source at a discrepant rate can simply be estimated from the difference between the best grid rate discrepant rates, and the centroid of the source at the best grid rate. The predictive filter[13] then considered all rates, searching for sources with lower S/N that are found near the predicted positions with a tolerance of 1.5 pixel radius, and any possible sources were collected and identified as a cluster of detections of a single source. Those detections were removed and the process repeated on the next remaining highest likelihood source, and so on until all candidates had been considered. Any source that could not be associated with a cluster was rejected from further consideration. While it is possible that the predictive filter could misassociate valid detections of nearby clouds, this was a very rare occurrence because the probability of two objects sharing very similar positions and velocities that could trick the filter was extremely low; a manual check showed that this never occurred, even for our implant list.

Finally, the peak S/N source in each cluster then underwent a filter that varied the position of a source to ±3 pixels around its highest likelihood point, akin to the brightness variation filter discussed above. Then, any source with a likelihood greater than 5 was kept. These are the shift–stack candidates, and still have a bogus:good ratio of more than 100:1.

4.2. Machine Learning Vetting

While the shift-and-stack method provides a fast and efficient way of searching over moving object candidates across multiple images, it still requires a significant amount of human effort for visually vetting real moving detections, even after the image subtraction process. To reduce the number of such false-positive outputs from shift-and-stack, we integrated machine learning techniques to perform an initial filtering of bad sources from the candidate list.

We explored different types of neural network architecture, each trained to perform a binary classification task, determining whether a source should be accepted as good or rejected as bad. Our experiments included standard convolutional neural networks (CNNs) with a range of complexities and depths to evaluate how architectural choices influenced classification performance. While those models delivered reasonable results, we achieved a significant improvement in performance when using an ensemble of residual networks (ResNets; K. He et al. 2016). The ResNet architecture we

[13] This predictive filter is what gives the shift–stack software package its name, `pkbmod`, which is available here https://github.com/fraserw/pkbmod.

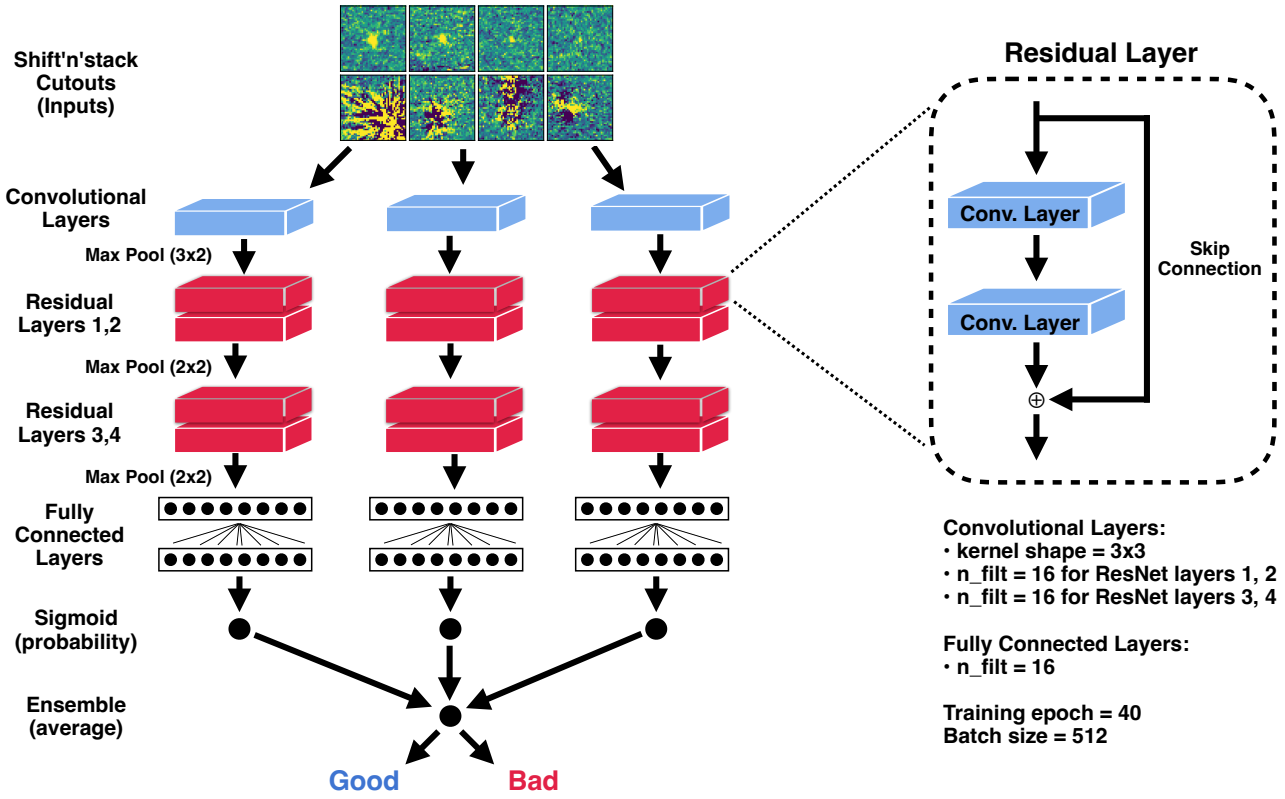


Figure 6. The network adopted for this project is an ensemble of three ResNets. Each *branch* of the ensemble has four residual layers utilizing 16 filters, represented in red blocks. Each red block contains two convolutional layers. The outputs from the fourth residual layer are fed into two fully connected layers, which ultimately lead to a sigmoid probability layer. The sigmoid outputs from all three models are then averaged to generate a final probability score for binary classification. For a detailed explanation of ResNet operations, we refer the reader to K. He et al. (2016).

used here is similar to the one employed in W. C. Fraser et al. (2024), which constructs a *master* network made up of multiple ensembles (or *branches*) of simpler networks. Our chosen ResNet model consistently outperformed all previous CNN-based models in terms of detection depth and peak detection efficiency, while also significantly reducing the number of trainable parameters.

Our adopted ensemble consists of three identical multilayer ResNets (see Figure 6), each independently initialized and trained on the same dataset. By combining multiple models into one ensemble, we improve the network's predictive performance, robustness, and accuracy. This approach is based on the idea that a group of models working together can often make better decisions or predictions than a single model. The final prediction probability of the ensemble is calculated by averaging the output probabilities from each individual ResNet.

We trained all our neural networks using supervised learning, where the model learns to identify features in manually labeled training images. Developing an effective training procedure was a challenging and iterative task, and the approach that ultimately proved successful is described as follows. A shift-and-stack detected source was labeled as *good* if it was located within 1 pixel of a known implanted source and its rate of motion matched the implant's to within $0\farcs3\,\mathrm{hr}^{-1}$. All other sources that did not meet this criterion were labeled as *bad* (see Figure 7). However, the shift-and-stack output was overwhelmingly dominated by junk/bad detections, with a ratio of approximately 350:1 compared to good sources—far too imbalanced for effective training. To address this, we built our training dataset by including all good detections and a randomly selected subset of bad detections at a ratio of 1.5:1 relative to the good sources. We find that increasing the bad-to-good ratio further would cause the model to become overly biased toward rejecting candidates, significantly reducing its ability to correctly identify true positives. Additionally, we applied sample weights of 1 for good sources and 0.8 for bad candidates, which helped achieve more balanced and accurate network performance. Furthermore, to increase the sample size and introduce necessary variability for training, we applied several augmentation techniques to our images. In particular, each source image was flipped horizontally and vertically, rotated 90° clockwise and counterclockwise, and duplicated with a one-pixel shift in four directions: left, right, up, and down. Such augmentations would help our model to learn to recognize features in our images regardless of orientation. Our sample size increased to 56,850 after augmentation, where 95% were used for training, and the remaining 5% were used for validation.

We explored multiple hyperparameter settings and ultimately settled on the configuration shown in Figure 6. Then, using 10 different training sets (each using the same set of good sources but a newly randomized selection of bad sources, as described above), we trained 10 different iterations of that network. Among the 10 resulting models, we selected the one that achieved the best performance, defined by the faintest detection depth and highest peak detection efficiency. Once the optimal model was chosen, we used it to categorize all outputs from our shift-and-stack

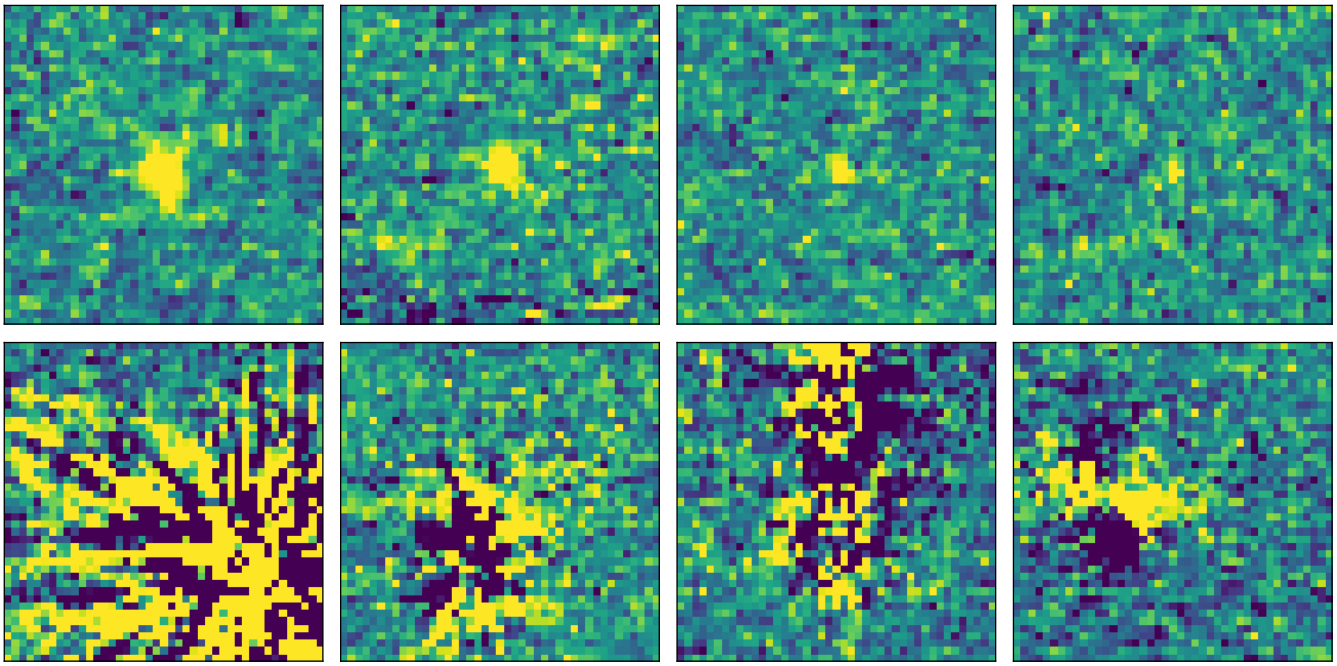


Figure 7. The upper panel displays normalized examples of source stamps (with normalized pixel values) classified as *good* by our selection process, while the lower panel shows those labeled as *bad*. Objects with shift-and-stack shifts that closely match their movement appear as nearly round sources in the stacked images. In contrast, bad sources are typically a result of poorly matched shift-and-stack shifts or residual artifacts left by very bright sources after image subtraction.

pipeline. We set the probability acceptance threshold to 0.65, a value we manually determined to balance sensitivity and precision before predictions became overly contaminated by false positives. Appendix A provides a graphical summary of network and threshold selection.

The performance of the network ensemble was impressive. Our shift-and-stack pipeline produced over a million detections—some artificial implants, some real sources, but the majority were junk. Our chosen network successfully filtered this to roughly 50,000 candidate sources across 20 NIRCam tiles and three epochs, though this came with a loss of about ∼0.2 mag in total depth compared to the original shift-and-stack results. Without this significant reduction in false positives, the subsequent vetting process would have been impractical. The sources identified as good were subsequently subjected to scene modeling to assess whether their image data are consistent with the profile of a moving point source.

Lastly, a primary concern with using machine learning was overfitting, a common issue where a model recognizes specific details of the training data instead of learning the underlying patterns. In this case, the model will label a source used in training more confidently than it would a source not used in training. To investigate this, we created a duplicate of our ensemble and trained each version on completely separate datasets, then validated each model using the dataset on which the other was trained. This cross-validation approach allowed us to assess whether the model's behavior and performance remained consistent when exposed to data outside their training distribution. We found that both ensembles produced similar results in terms of similar detection depth and peak efficiency, suggesting that our chosen network was effectively learning the general features of the data and that overfitting was not a significant issue in our setup.

4.3. Scene Modeling—Single Epoch

4.3.1. Scene-modeling Technique

Once the ResNet classification has identified a candidate TNO, we perform scene-modeling photometry (SMP) on the original (unregistered) images to confirm that the image data are consistent with a model of a moving point source atop a static sky background. The SMP process also yields optimal estimates of the flux and position versus time of the moving source, and the uncertainties on these quantities.

The SMP procedure is similar to that used by P. H. Bernardinelli et al. (2023). For a candidate single-epoch track in some region of the search field, we locate all exposures that covered this region and extract similar "postage stamps" of ≈25 × 25 pixels from each exposure centered on the nominal center of the TNO track proposed by the shift-and-stack software. Using j as the index over exposures,[14] and using (m, n) to represent the pixel indices within the exposure, we model the value in the image as

$$\hat{f}_{\mathrm{jmn}} = b_j + \sum_{\mu,\nu} f_{\mu\nu} P_j(u_{\mathrm{jmn}} - u_{\mu\nu},\, v_{\mathrm{jmn}} - v_{\mu\nu}) + f_j^T P_j[u_{\mathrm{jmn}} - u^T(t_j),\, v_{\mathrm{jmn}} - v^T(t_j)]. \quad (1)$$

In this equation, (u, v) are sky-plane coordinates—all coordinates are taken in a gnomonic projection about $(\alpha, \delta) = (209.^{\circ}40, -10.^{\circ}87)$. The pixels of the image have central coordinates $(u_{\mathrm{jmn}}, v_{\mathrm{jmn}})$, known from the WCS of each exposure. Each stamp also has a known PSF $P_j(u, v)$, and an unknown constant background level b_j. We presume that the sky consists of static point sources on a grid indexed by (μ, ν)

[14] In this context, "exposure" means one count-rate image from a single NIRCam integration.

at sky positions ($u_{\mu\nu}$, $v_{\mu\nu}$). The fluxes $f_{\mu\nu}$ of the background are free parameters of the model.

The functions $u^T(t)$, $v^T(t)$ are the candidate sky motion of the target TNO, and the fluxes f_j^T are its observed light curve. When fitting a model to a single observing epoch, we assume linear motion:

$$u^T(t) = u_0 + \dot{u}(t - t_0)$$
$$v^T(t) = v_0 + \dot{v}(t - t_0). \quad (2)$$

We assign a reference time t_0 to the track, and then the parameters $\{u_0, v_0, \dot{u}, \dot{v}\}$, the position and rate, are free.

The SMP fits this model to the data, maximizing the log-likelihood of the data, which under an assumption of Gaussian uncertainties $\sigma_{\rm jmn}$ on each pixel datum, becomes minimization of the usual

$$\chi^2 \equiv \sum_{\rm jmn} \left(\frac{f_{\rm jmn} - \hat{f}_{\rm jmn}}{\sigma_{\rm jmn}} \right)^2. \quad (3)$$

Pixels with error flags set are omitted from the calculation. The free parameters in this model include the exposure backgrounds $\{b_j\}$, the static scene $\{f_{\mu\nu}\}$, the target light curve $\{f_j^T\}$, and the ephemeris parameters $\{u_0, v_0, \dot{u}, \dot{v}\}$. All except the last group enter the model linearly, and hence their values that minimize χ^2, and their uncertainties, can be found algebraically for a given choice of ephemeris. We minimize χ^2 over the (nonlinear) ephemeris parameters using a Newton–Raphson iteration, and derive a 4×4 covariance matrix for the ephemeris parameters by linearizing the model.

In practice, we usually derive the scene for a track candidate in epoch 1 by using the exposures from epochs 2 and 3 (similarly for candidates in epochs 2 and 3). Then we fit the stamps from epoch 1 while holding the scene $\{f_{\mu\nu}\}$ fixed while varying the background and TNO models. The $\mu\nu$ grid is square (on the sky) with a pitch of one NIRCam pixel. Some nearly degenerate modes of the scene solution are nulled.

PSF models are obtained from the `WebbPSF` package provided by STScI.[15] The WCS models are described in Section 3.3.

4.3.2. Quality Control and Cuts: Single-epoch Tracks

The SMP analysis yields the optimal values and uncertainties on the TNO fluxes f_j^T and the astrometry, and in addition several quantities useful for discriminating true TNO tracks from spurious shift-and-stack triggers, such as noise peaks, unmasked detector defects, image persistence in the detector, and subtraction errors. The main characteristic that we will use to mark a track as valid versus spurious is that a TNO will have undetectable variation in flux during the three integrations making up one of the two dithered exposures in the epoch. For each dither with exposures $j = \{1, 2, 3\}$, we use the fluxes f_j and flux uncertainties $\sigma_j(f)$ to calculate

$$\bar{f} \equiv \frac{\sum_j f_j/\sigma_j^2(f)}{\sum_j 1/\sigma_j^2(f)}, \quad (4)$$

$$\sigma_f^2 \equiv \frac{1}{\sum_j 1/\sigma_j^2(f)}, \quad (5)$$

$$\chi_\nu^2(f) \equiv \frac{1}{2}\sum_j (f_j - \bar{f})^2/\sigma_j^2(f), \quad (6)$$

$$S/N \equiv \frac{\bar{f}}{\sigma_f}. \quad (7)$$

Figure 8 illustrates how we use the resultant S/N and $\chi_\nu^2(f)$ values to eliminate most of the spurious tracks with minimal loss of real TNOs' tracks. We demand that both dithers yield detections of the source with S/N > 2.5 using the dithers' common rate of motion, i.e., the source cannot disappear. We also demand that the reduced χ^2 of the three fluxes indicate loosely consistent flux during each 10 minute exposure, $\chi_\nu^2(f) < 7$. The left side of the figure shows that this encompasses nearly all tracks that are position-matched to an implanted object—and some of the ones that fall outside the selection region are seen upon visual inspection to be coincidences of an implant near a noise or artifact track. The right-hand plot shows that the number of spurious tracks increases exponentially at low S/N even though most of the brighter ones can be excluded by the $\chi_\nu^2(f)$ criterion (red tinged). Please note that we never examined any plots with the implanted objects removed until *after* the selections were made. We wanted to remain blind to the number of detectable real TNOs while making analysis choices, so we only examined plots with the implants and observed tracks mixed together.

One diagnostic that we calculate but do *not* use to cull the sample is χ_5^2, defined to be the total χ^2 of the data relative to the scene model, summed over all the valid pixels in all integrations contained within a 5×5 pixel box centered on the best-fit TNO position. A high value of χ_5^2 relative to the number of included pixels indicates that the data are a poor fit to the scene model, suggesting an artifact. It is also true, however, that bright TNOs are expected to have $\chi^2 > 1$ per pixel; the PSF models are not perfect, and the CR rejection algorithms for the NIRCam up-the-ramp sampling can be spuriously triggered by a bright moving source, biasing the output rate images. We also might obtain high χ^2 from binary TNOs. We therefore do not exclude tracks based on their χ_5^2 values. Instead, we use this as a collective indicator that our models and uncertainties are sensible, finding that $\langle \chi_5^2 \rangle$ per pixel is indeed near unity for tracks with S/N $\lesssim$ 30, rising for TNOs brighter than this. In Figure 8, one can see some implants at high S/N failing the cut on $\chi_\nu^2(f)$, likely caused by PSF mismatches and spurious triggers of the CR clipping algorithm. We therefore apply the $\chi_\nu^2(f)$ cut only for S/N < 30. Since sources this bright are rare, and brighter than a pure noise fluctuation can produce, and easily classified as real/spurious by eye, we can remove this cut and still obtain a highly reliable final catalog.

Figure 9 plots the distribution of the total S/N of the 14,000 unique tracks that pass these two cuts (total S/N being the quadrature sum of the two dithered exposures' S/N values). The exponential increase at S/N $\lesssim$ 7.5 is the behavior one expects from noise detections over a wide search area. We have chosen cuts for `kbmod`, the ResNet classifier, and track S/N that allow our sample to be dominated by noise detections

[15] https://github.com/spacetelescope/webbpsf

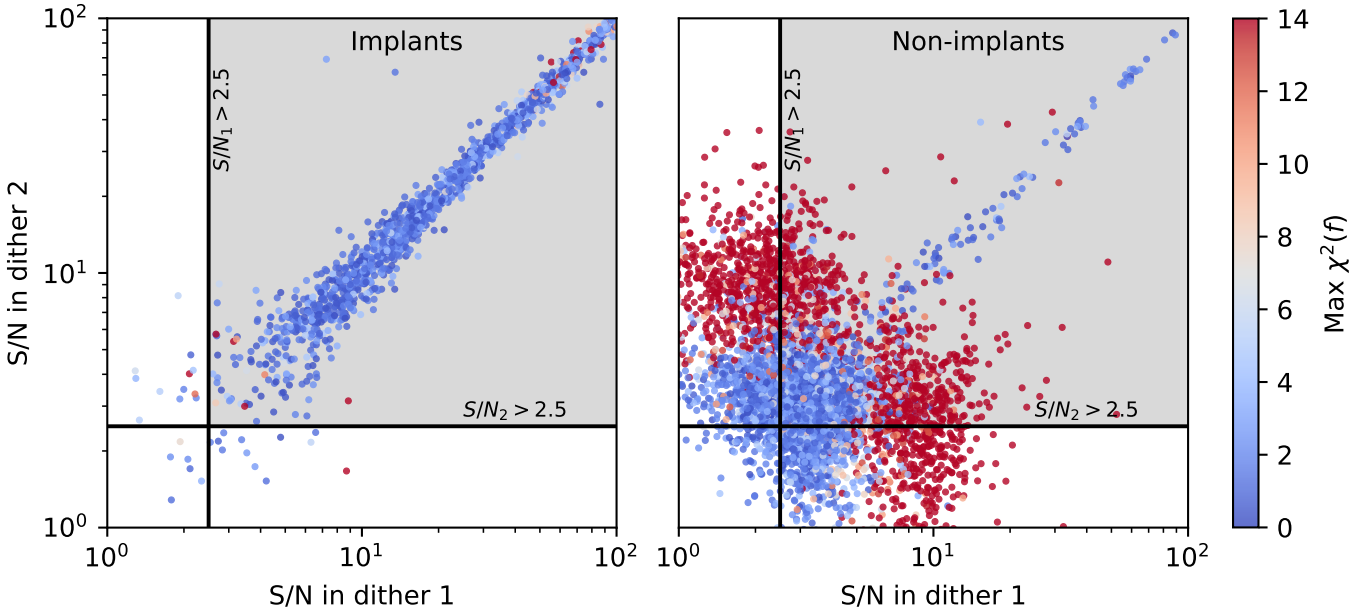


Figure 8. The criteria used to select "good" single-epoch tracks are illustrated. The axes of each panel are the total S/N of the flux detection (Equation 7) in the two dithered exposures of the track's epoch. Each point is the result of a scene-model fit to one KBMOD detection of a moving source. The color of the point codes the worse of the internal flux consistency measures $\chi^2_\nu(f)$ of the two dithers. Our cuts are chosen from the distribution of the values for the implanted objects, shown at left; we require S/N > 2.5 in *both* dithers (the gray region), and also $\chi^2_\nu(f) < 7$ in *both* dithers, i.e., only points with a blue tinge. The right-hand side shows the distribution of the nonimplanted, potentially real objects. We have subsampled the tracks with S/N < 10 to avoid overcrowding the figure; nonetheless, the flood of noise or artifact tracks at S/N < 5, but with acceptable $\chi^2_\nu(f)$, is apparent.

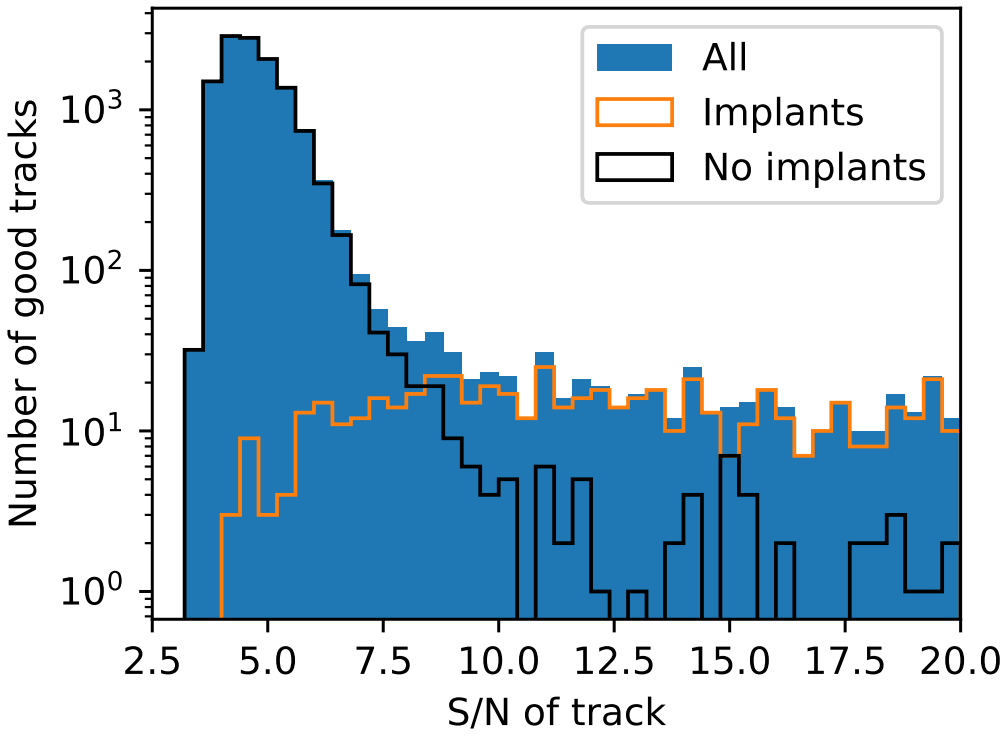


Figure 9. We plot the distributions of total S/N for all tracks that pass the cuts from Section 4.3.2 on the per-dither S/N and $\chi^2_\nu(f)$. These are shown for the implants, the nonimplanted tracks, and the total. Note the logarithmic y-axis. The exponential increase in nonimplanted tracks for S/N < 10 indicates the onset of noise detections, which dominate the total track counts for S/N < 7.5

at the single-epoch track level. The goal is to reliably detect real TNOs to the faintest possible flux. The spurious tracks will be weeded out by their failure to link with tracks from the other two epochs in a plausible TNO orbit, as investigated later in this section.

4.3.3. Completeness

The probability with which a TNO will be detected as a good track during a single epoch, if it falls on a detector during both dithers, is quantified using the implants. It is well fit by

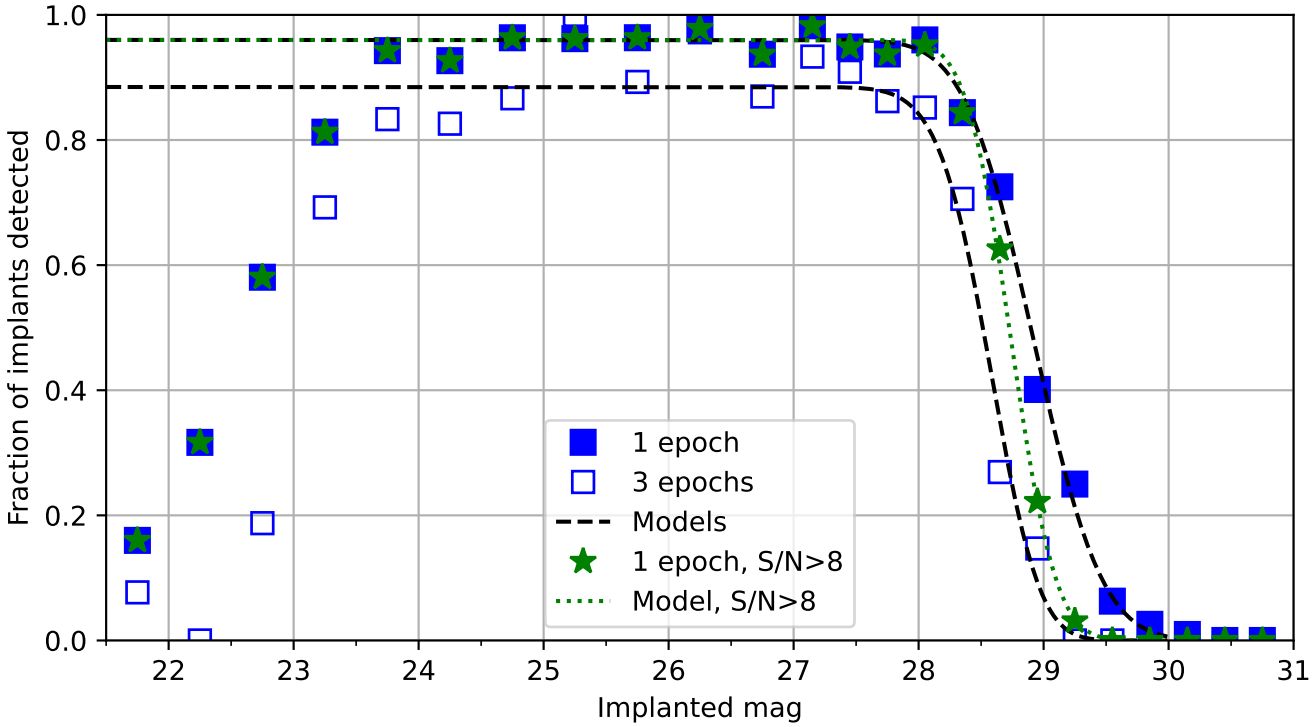


Figure 10. The filled blue squares plot the fraction of objects implanted onto valid regions of the NIRCam SW detectors that yield a fitted track that passes the quality cuts described in Section 4.3.2, as a function of injected magnitude. The upper dashed black line plots the completeness function $p_{\rm det}$ from Equation (8) with parameters that maximize the likelihood of the implant results. The analytic function is seen to be a good representation of the implant results. The lower black curve plots the cube of the upper completeness function, which is the expectation for independent detection probabilities in all three epochs, and agrees with the implant results plotted as open blue squares. The green stars and the dotted green line are the measured implant recovery rate and fitted model when a cut of S/N > 8 is imposed on the tracks, as is done in creating sample B of TNOs in Section 5.5. Lastly, we stress here that the decline in efficiency at the bright-end results from the tuning applied during the shift-and-stack stage of the search, rather than from overtraining of the ResNet.

the functional form

$$p_{\rm det}(m) = \frac{p_0}{2}\,\mathrm{erfc}\left(\frac{m - m_0}{w}\right) \tag{8}$$

with the bright-end efficiency $p_0 = 0.96$, the magnitude of half that efficiency at $m_0 = 28.92$, and transition width $w = 0.61$ mag. Figure 10 plots this curve, along with binned rates of detection for the implants. This suggests that about 4% of sources are lost from overlaps with bright sources or defects on the detectors. It should be noted that the PSF model is most likely responsible for the decrease in detection efficiency at the brightest end of our implanted magnitude range. That is, inaccuracies in the PSF combined with the tolerances in the predictive filter tended to reject the brightest implants at a slightly worse rate than for fainter sources for which the tolerances were tuned.

We also check the probability that an implant that falls on a detector in all of the three epochs is detected as a valid track in all three epochs. The binned success fractions are shown as open symbols in Figure 10, and are well fit by the cube of the single-epoch $p_{\rm det}$ function in Equation (8), which is what we expect if the detection process is statistically independent in each epoch.

In Section 5.5, we will create a sample of detections constrained to obtain S/N $\geqslant 8$ in single epochs. The green stars in Figure 10 plot the measured recovery efficiency of implants subject to the condition. The recovery rate is well fit by Equation (8), with parameters $p_0 = 0.96$, $m_0 = 28.74$, and $w = 0.40$ mag.

4.4. Lightcurve Bias

TNOs with larger amplitude variations are more likely to be missed at certain rotational phases, even if their mean brightness is above the survey's limiting magnitude. To test whether this affects our survey, we simulate 1000 objects of varying rotational periods, peak-to-peak amplitudes Δm, and magnitudes. For each simulated object, we calculate the three-epoch efficiency value and plot it as a function of their lightcurve mean magnitude and peak-to-peak amplitude (see left of Figure 11). Fainter objects with high Δm (red circles) and whose cube efficiency is high were likely recovered when they are at brighter phases. In parallel, we also calculate the fraction of recovered implants over our three-epoch survey as a function of their amplitude, as seen on the right of Figure 11. Typically, we would expect a drop in the recovered fraction for objects with higher amplitudes, as their brightness occasionally falls below the detection threshold. However, the recovery fraction is notably high for the bins at 1.05, 1.20, and 1.35, although with large uncertainties due to the very small number of implanted sources in these bins. Consequently, we do not find any variation of our survey completeness with lightcurve amplitude; however, the insufficient number density of our implanted objects limits us from robustly quantifying this effect in our survey.

4.4.1. Decoupled Errors for Linking

While SMP yields the best-fit position (u_0, v_0) and rate $(\dot{u}, \dot{v})$ for a single TNO track, with their full covariance matrix, the linking software is simplified if we convert these four variables

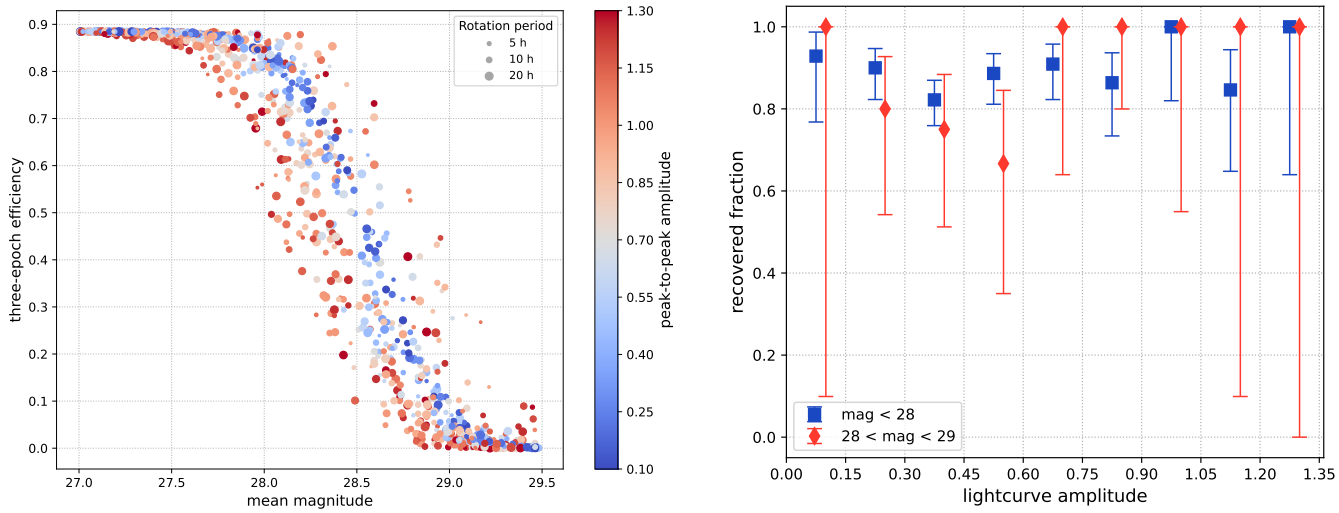


Figure 11. Left: three-epoch detection efficiency for simulated objects with a mean magnitude greater than 27 mag. The data points are color coded according to their peak-to-peak brightness variation, Δm. The sizes of the markers are proportional to the objects' spin rates, which span from 2 to 20 hr. Right: fraction of recovered implants over three epochs for objects with implant magnitudes between 28 and 29 mag (green diamonds), where the detection efficiency begins to decline. For reference, blue squares show the recovery fraction for implants with magnitudes <28 mag. Lightcurve amplitude is binned in 0.15 mag intervals, and error bars indicate 1σ Poisson uncertainties on the number of missed objects in each bin.

into two "pseudodetections" at times t_1, t_2 with assigned positions (u_1, v_1) and (u_2, v_2). These positions are chosen so as to yield the same position and rate originally found, and the times are selected such that the covariance between (u_1, v_1) and (u_2, v_2) is near zero. Each epoch is thus converted into two pseudo-observations, each with its own position errors, that can be treated as statistically independent measurements by the linking code, while carrying the same information as the position/rate format.

4.5. Cross-epoch Linking

At this stage, we have pairs of pseudodetections that can be treated as independent observations and carry the position/velocity information from a single track. The challenge is to determine which tracks, observed at distinct epochs, correspond to the same real or implanted object. This is known as *linking*.

The observed angular coordinates of the candidate TNO in the sky plane are given by

$$u^T(t) = \frac{x(t') - x_E(t)}{z(t') - z_E(t)}$$
$$v^T(t) = \frac{y(t') - y_E(t)}{z(t') - z_E(t)}, \quad (9)$$

where $t' = t - \Delta t$, and Δt is the light travel time from the object to the observer.

Each track's pseudo-observations provide four quantities (the two sky positions). However, a full Keplerian orbit requires six parameters. The two missing elements are effectively the object's distance from the observer and its line-of-sight velocity at a reference time. We adopt the `HelioLinC` approach (M. J. Holman et al. 2018), which involves asserting values for the two unknown quantities, computing the corresponding orbit for each track, and then identifying clusters of tracks from distinct epochs that have yielded similar orbits. When the assumed values are correct, the resulting orbits for tracks that correspond to the same object should differ only due to the measurement uncertainties of the observations.

We adopt a version of the coordinate system of G. Bernstein & B. Khushalani (2000). The z-axis points in the direction of (R.A., decl.)=($209.^\circ4$, $-10.^\circ87$), approximately the center of our search region. The x- and y-axes lie in the sky plane normal to z. Rather than following the ecliptic, we align the x-axis with the direction of increasing R.A., and the y-axis then points in the direction of increasing decl. The reference time is set to JD 2459974.5 (TDB), which is near the midpoint of the observation period.

For each pseudo-detection, we look up the position of JWST at its time of observation and transform it into our coordinate system. To account for the limitations of the PSF model for bright sources, we add $0.''0005$ in quadrature to their positional uncertainties. Then, we use the parameterization of G. Bernstein & B. Khushalani (2000), based on the components of the position and velocity of the target at the reference time: $\alpha = x_0/z_0$, $\beta = y_0/z_0$, $\gamma = 1/z_0$, $\dot\alpha = \dot x_0/z_0$, $\dot\beta = \dot y_0/z_0$, and $\dot\gamma = \dot z_0/z_0$. With these, α and β are the components of the angular position of the object at the reference time, $\dot\alpha$ and $\dot\beta$ are angular rates of motion in the inertial coordinate system, γ is a measure of distance to the object, and $\dot\gamma$ is a scaled radial velocity.

In terms of these parameters, the observation θ_x is given by:

$$\theta_x(t) = \frac{\alpha + \dot\alpha t' + \gamma g_x(t') - \gamma x_E(t)}{1 + \dot\gamma t' + \gamma g_z(t') - \gamma z_E(t)}, \quad (10)$$

where $t' \approx t - \frac{1}{c\gamma}$ is the light-time corrected time of the observation. There is a corresponding equation for the fitted observation θ_y.

Equation (10) can be rearranged to yield a simple expression for the linear motion of the object:

$$\alpha + \dot\alpha t' = \theta_x[1 + \dot\gamma t' + \gamma g_z(t') - \gamma z_E(t)] - \gamma g_x(t') + \gamma x_E(t), \quad (11)$$

where θ_x and θ_y are observed quantities, and the observatory position (x_E, y_E, z_E) is known precisely. We note that in Equation (11) the transverse components of the gravitational perturbation, $g_x(t')$ and $g_y(t')$, are much smaller than $g_z(t')$ and can be ignored.

Finally, to investigate possible orbital configurations, first, we assume a constant acceleration due to the total mass of the solar system in the direction of the barycenter. This is an excellent approximation for the time span of these observations, as the acceleration is very small (G. Bernstein & B. Khushalani 2000). Then, we loop over pairs of γ and $\dot{\gamma}$, where γ is uniformly spaced between 0 and 0.06 au^{-1} with 721 values, and $\dot{\gamma}$ is uniformly spaced between -10^{-4} rad day^{-1} and 10^{-4} rad day^{-1} with seven values. This range of values encompasses bound orbits from inside the orbit of Uranus to infinity. We established the spacing of the values empirically.

This rapid algebraic search for valid orbit fits is applied to all potential triplets of pseudo-observation pairs from three distinct JWST observing epochs (`HelioLinC` uses clustering algorithms that do not actually require examining every distinct triplet). From the ∼14,000 single-epoch tracks passing the quality cuts in Section 4.3.2, we find ∼21,000 unique three-epoch triplets that are consistent with a bound orbit in the sense of having rms residuals $\leqslant 0\farcs2$ between the measured positions and the positions predicted by the orbit model. The vast majority of these include at least one track that yielded $\mathrm{S/N} < 6$ across its JWST epoch. The $\leqslant 0\farcs2$ cut on orbit-fitting rms is intentionally very loose and admits many false-positive linkages. The next section describes how these candidates are culled to eliminate false positives at high confidence while retaining as many real TNOs as possible.

There are 417 triplets of implanted tracks detected (some implanted objects are detected more than once in a given epoch, leading to multiple ways to link one per epoch). Of these, the linking process described herein finds all but two. In fact, these two cases are very faint implants, which we suspect are actually being mismatched to nearby spurious detections, and thus should *not* yield good links. We conclude that the linking process has negligible false negatives.

5. Orbit Refinement and Selection

5.1. Three-epoch Linkages

The previous section's linkage procedure produces a list of candidate TNO orbits, each consisting of one track in each of the three epochs. This section describes how each of these candidate orbits is refined, and selection cuts are made to exclude spurious linkages while retaining real sources. This procedure parallels the treatment of single-epoch tracks in Section 4.3; first, each orbit candidate is subjected to scene modeling. Next, quality cuts are defined based on the results for implanted sources, such that the false-negative rate for implanted objects is kept minimal while rejecting all readily identifiable spurious linkages. Finally, the full population of three-epoch orbit candidates—which is a mixture of real TNOs, implanted TNOs, and spurious detections, or mixtures of these types, with the investigators blind to which is which—is passed through the quality cuts. A rapid upturn in the number of accepted triplets at the faint end would be the telltale sign of false-positive detections, and we find no such upturn. We end up with a list of validated three-epoch TNO detections, and can unblind the results by removing implanted sources from this list.

The SMP for full orbits again models each integration's image (three integrations per dither × two dithers per epoch × three epochs) as the sum of a time-invariant background scene, plus a point-source TNO with time-varying flux, following an ephemeris $[u^T(t),\ v^T(v)]$. The main difference between the single-epoch SMP and the full-orbit SMP is that we replace the linear apparent motion of Equations (2) with an ephemeris parameterized by the initial ICRS barycentric Cartesian state vector ($\boldsymbol{x}_0$, $\boldsymbol{v}_0$) of the TNO. The TNO motion is integrated under the gravity of the Sun and giant planets (assumed to be constant over our 10 day observing period), and the observatory's state vector is retrieved from JPL Horizons to permit calculation of apparent coordinates versus time of observation using Equations (9). The orbit is fit to exposures of all three epochs simultaneously, and yields a maximum-likelihood initial state vector and a covariance matrix for its six elements. A separate TNO flux is fit to each NIRCam integration. Another difference from the single-epoch SMP process is that there are three distinct static background images; the background for the location of the TNO in epoch j is ascertained by using images of the same sky region in the other two epochs. A total of 54 postage stamps are thus extracted and fit for each candidate TNO orbit.

Figure 12 illustrates the result of the full three-epoch SMP process for one newly discovered TNO. The auxiliary files available with this paper contain similar diagrams for all of the three-epoch TNOs, as well as the numerical arrays comprising all relevant image cutouts and models (see Appendix C).

The outputs of the multiepoch SMP are the estimated fluxes f_j at each of the 18 integrations and their uncertainties $\sigma_j(f)$, plus a best-fit ICRS barycentric state vector at the reference time, and its covariance matrix. The main diagnostic criteria for distinguishing real from spurious linkages are whether the fitted orbit is indeed the best description of the data in the three original epochs. For spurious linkages, the three individual-epoch fits are likely to better fit the pixel data than constraining the TNO to follow a physical orbit across three epochs.

Three quantities are used to select genuine TNOs from among the ≈21,000 three-epoch linkages returned by the algorithms of Section 4.5. To derive the first, we calculate the flux at each epoch using Equation (4) to average over the epoch's six integrations, then convert this flux into an *AB* magnitude using the calibrations provided by the Space Telescope Science Institute. We then define $\Delta m = m_{\mathrm{orbit}} - m_{\mathrm{track}}$. The magnitude derived from fluxes returned by SMP for that epoch when all three epochs are fit simultaneously to an orbit is m_{orbit}, and m_{track} is the magnitude measured by SMP when fitting a linear motion to just the single track (Section 4.3). A value $\Delta m > 0$ will result if the PSF being used by SMP for the full-orbit fit is not well centered on the object's flux, which is a sign that the three tracks do not actually arise from a TNO on a Keplerian orbit. We take $\max(\Delta m)$ over the three linked epochs as one quality indicator.

The second selection asks whether the apparent rate of linear motion $\boldsymbol{r}_{\mathrm{track}} = (\dot{u}, \dot{v})$ fit to each individual track via Equation (2) agrees with the rate $\boldsymbol{r}_{\mathrm{orbit}}$ implied by the orbit derived as best fit in the three-epoch SMP. The SMP for each epoch returns a 2 × 2 covariance matrix C_r for the components of $\boldsymbol{r}_{\mathrm{track}}$. We take the maximum over the three epochs j of the statistical significance of the disagreement between the single-epoch rate of motion and the orbit fit's rate of motion:

$$\Delta\chi^2_\nu(\boldsymbol{r}) \equiv \max_j \Big[(\boldsymbol{r}_{\mathrm{track},j} - \boldsymbol{r}_{\mathrm{orbit},j})^T \times C^{-1}_{\boldsymbol{r},j} (\boldsymbol{r}_{\mathrm{track},j} - \boldsymbol{r}_{\mathrm{orbit},j})/2 \Big]. \quad (12)$$

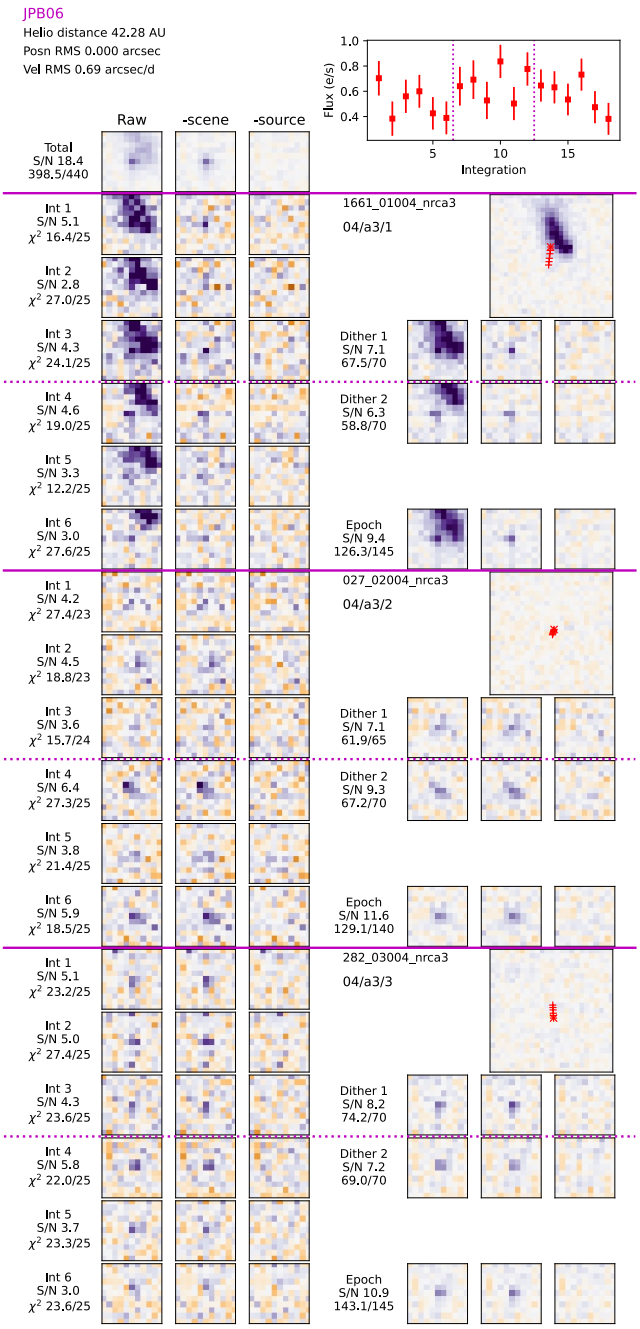


Figure 12. Illustration of the orbit-level SMP fit to the newly discovered TNO JPB06 at $m_{AB} = 28.6$ mag. Each row shows a set of three cutouts from the NIRCam SW data spanning 0.″34. The left is the calibrated image, the center subtracts the model for the background, and the right also subtracts the model of the TNO moving along the best-fit orbit. Ideally, the right-hand images are pure noise. Between each pair of purple lines, the images from one epoch's six individual integrations are stacked at the left. At the right are (top to bottom) the background scene derived for the region; the means of the three integrations in each of the two dithered exposures; and the mean of all six exposures, i.e., "shift-and-stack" images. The S/N of this source, and the χ^2/DOF of the model fit in the 5 × 5 pixels centered on the TNO, are shown for each individual integration and stack. At the top left are the results of averaging all 18 integrations of the target, and at the top right is a plot of the fitted flux versus integration number. The accuracy of the background+source model is apparent, as is the gain in detection significance as more exposures are combined.

A high $\Delta\chi^2_\nu(\boldsymbol{r})$ value implies that the motion observed between the three epochs does not agree with the motion observed within each epoch, contrary to behavior of a real TNO.

The final diagnostic property calculated for each orbit is $\min_j(S/N_j)$, the minimum among epochs j of the S/N of detection in that epoch. The S/N comes from Equation (7), using the f and σ_f values derived for the epoch's six integrations by the SMP fit to the full orbit. An orbital linkage to a noise peak, or other spurious track, is likely to yield a very low $\min(S/N)$.

Figure 13 plots the locations of the linked triplets in the plane of $\min(S/N)$ versus Δm in the top row, and $\min(S/N)$ versus $\Delta\chi^2_\nu(\boldsymbol{r})$ in the bottom row. At left, we plot the results for correct linkages of three epochs of tracks linked to the same implanted source. We see that if we define selection cuts

$$\begin{aligned} \min(S/N) &> 5 \\ \Delta m &< 0.4 \text{ mag} \\ \Delta\chi^2_\nu(\boldsymbol{r}) &< 7, \end{aligned} \tag{13}$$

then we successfully retain all of the 413 correct linkages of implants. The red circles are linkages that contain only one or two tracks linked to any implanted source, i.e., they are mislinks either between two different implants, or between implanted and nonimplanted sources, and hence are definitively mislinkages of accidentally aligned detections. We see that only one of the 7000 of these known mislinks passes the cuts of Equation (13) (open red circle within the gray regions), and this one is easily identified as a mislink upon visual examination of its SMP images.

Once these cuts are established at values that guarantee low false-negative rates for the implants, we are ready to apply them to the full population of three-epoch linkages and decide if these cuts also avoid false positives from spurious linkages. The plots in the right-hand column of Figure 13 show that 23 linkages are well separated from the many thousands of other linkages in these diagnostic plots and lie within the locus of known implants. These are our confidently detected three-epoch TNOs. Please note that the right-hand plots were not actually made until after the cuts in Equations (13) were decided by examining plots that marked the known mislinkages, but did not distinguish implanted triplets from nonimplanted triplets. Confidence that we have no false positives arises from the clear separation of the loci of known mislinks versus the correctly linked implants. After unblinding by removing all triplets including any implants, a similar separation into two loci is seen in the right-hand panel of Figure 13.

Note that our selection criteria do *not* require a valid orbit to have fluxes that match in any way across epochs or dithers, since that would induce a bias against highly variable TNOs. We also do not invoke any cuts on the reduced χ^2_ν of the SMP model's fit to the pixel data. We do calculate the χ^2_5 value per pixel of the full-orbit SMP models, but we do not invoke cuts on this quantity when making our TNO selection, for fear of eliminating potential binary TNOs or very bright sources from our sample. Figure 14 plots the χ^2_5-per-pixel values of all of the linkages that pass our cut, both implanted sources and the 23 real-sky triplets. We find the expected behaviors; low-S/N sources cluster around a value of ≈0.9, slightly below unity because the actual number of degrees of freedom of the fit is below the number of pixels. Also, the values rise for higher-S/N sources because of mismatches between the true

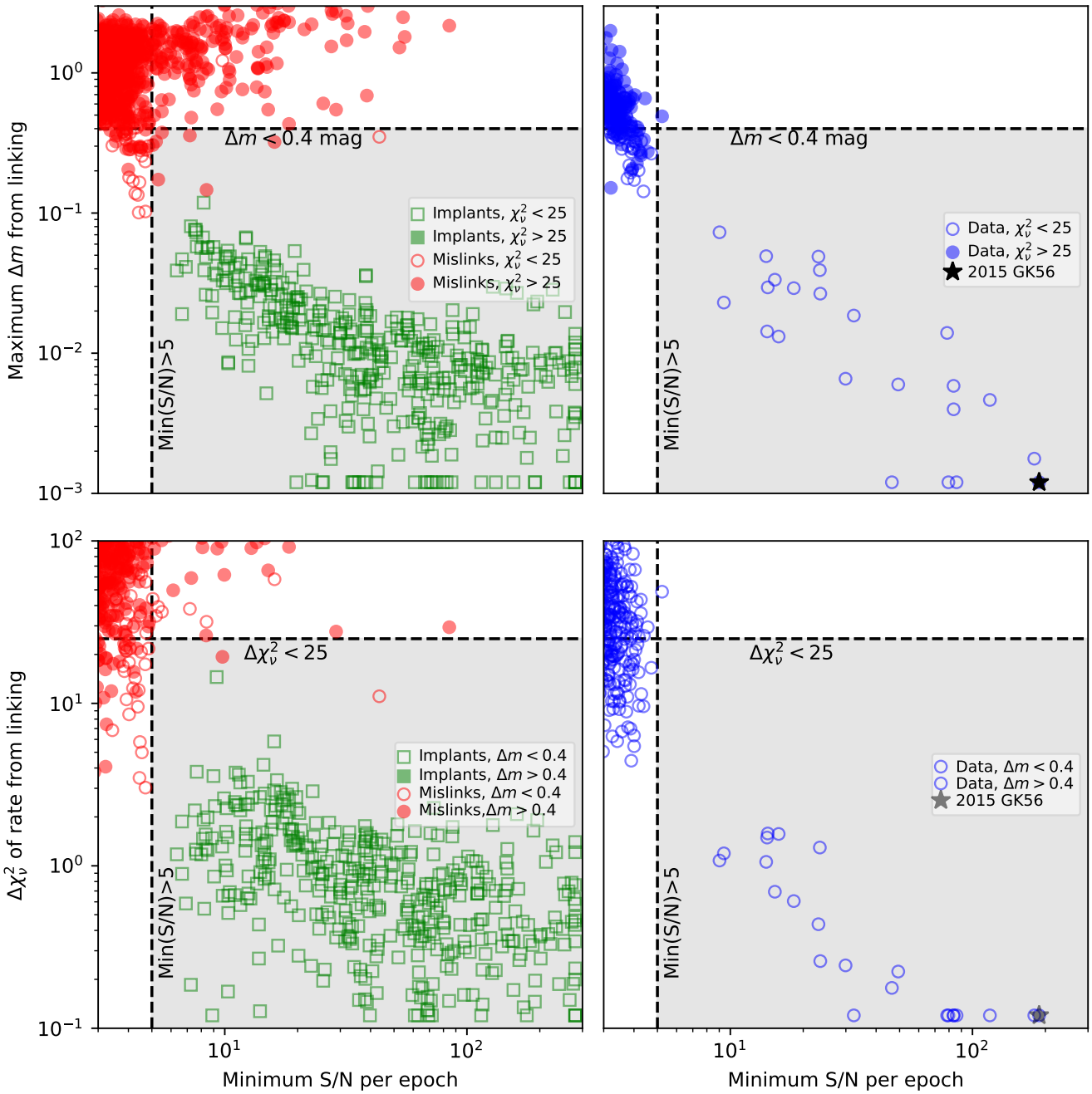


Figure 13. The top row of panels shows the positions of three-epoch linkages in the plane of diagnostic quantities $\min(S/N)$ (the lowest S/N of the putative TNO found among the three epochs) and Δm (the worst epoch's magnitude loss incurred by forcing a fit to an orbit) described in Section 5. The left panel plots candidate orbits that include three tracks from the same implanted source as the green squares, and tracks that include only one or two tracks from a given implant—i.e., spurious linkages—as the red circles. The right panel shows results for linkages that do not include any implanted sources. The symbols are filled if the selection $\Delta\chi^2_\nu(\boldsymbol{r}) < 25$ on the orbital versus track rates of motion is failed for any epoch. The gray region shows the cuts $\min(S/N) > 5$ and $\Delta m < 0.4$ that we use to separate genuine TNO linkages from spurious linkages. On the left, the combination of the three cuts is 100% effective in recovering correct linkages of implanted TNOs, with only one false positive of a mislink (open red circle inside the gray region), which is readily rejected by eyeball inspection. The right panel shows that 23 nonimplant linkages, including 2015 GK_{56}, are clearly isolated by these criteria out of the thousands of candidates. The bottom row shows the same process in the plane of $\min(S/N)$ versus $\Delta\chi^2_\nu(\boldsymbol{r})$, with symbols filled if they fail the Δm cut.

PSF and the model we used to fit. Notice that the bright implanted sources have higher χ^2_5 values than the real ones at the bright end (including 2015 GK_{56}), because the PSF model used during detection was closer to the true PSF than it was to the PSF assumed for implantation.

The 23 real-sky three-epoch linkages that pass selection cuts arise from 19 distinct TNOs. One of them is 2015 GK_{56}, and 18 are new discoveries. Four of the new TNOs were detected in two different tiles in one of the observing epochs, thus generating two successful three-epoch linkages per TNO. The 18 newly discovered TNOs have time-averaged fluxes corresponding to AB magnitudes of $24.5 \leqslant m_{\mathrm{F150W2}} \leqslant 28.6$. All 19 of these objects are listed in Table 1 along with their key photometric and dynamical properties.

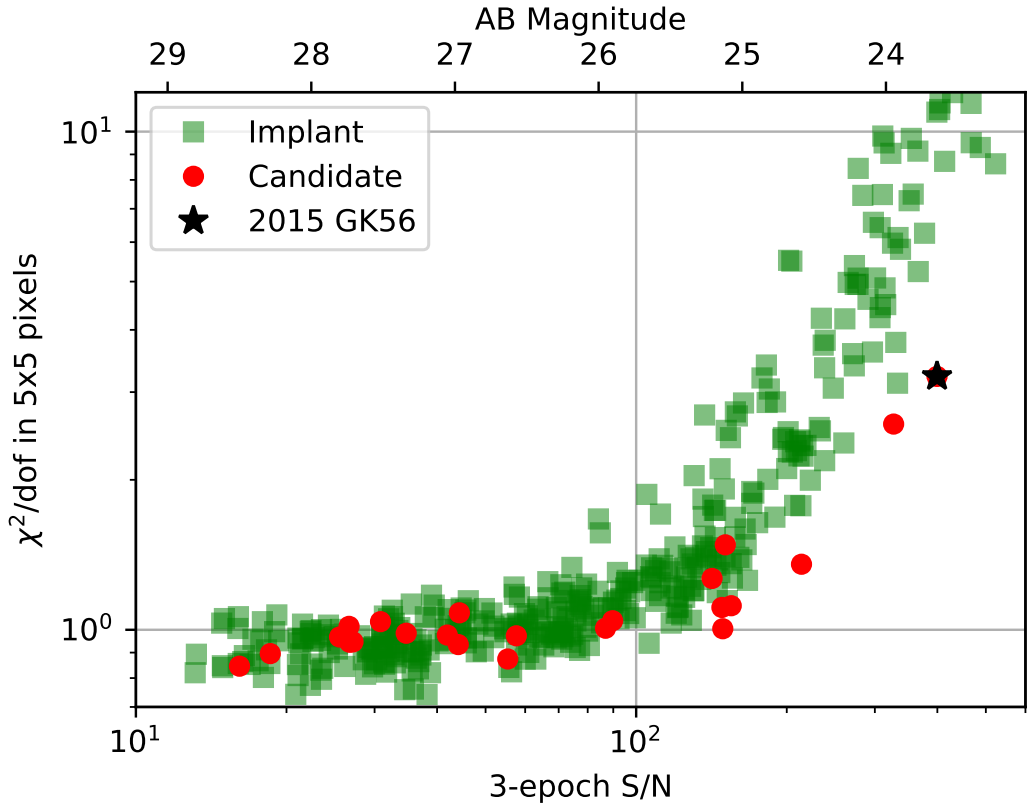


Figure 14. The χ^2 per pixel of the SMP fit to the pixel data is shown for all three-epoch orbit fits that pass the selection cuts in Equation (13). Only valid pixels in 5×5 pixel boxes around the source are used. Implanted objects are in green squares, newly discovered real-sky linkages are in red, and 2015 GK_{56} is the star. The statistic behaves as expected, with values slightly below unity at low S/N, rising for S/N $\gtrsim$ 30 sources for which the mismatch between the assumed PSF and the real or implanted PSF causes significant model error.

5.2. Two-epoch Linkages

We next search the list of nonimplanted single-epoch tracks for pairs that are consistent with a common TNO orbit. There are three motivations for this. First, it is expected that due to the gaps and edges of the NIRCam-imaged field, there should be a ratio of ≈0.3: 0.4: 1 TNOs that appear in 1, 2, and 3 epochs' images, respectively. This ratio was determined from the implanted orbits and is only mildly dependent on the barycentric distance d and inclination of the sources over the range $30 < d < 60$ au. The discovery of such objects is therefore expected and serves as a crude cross-check on our three-epoch linkage rates.

The second motivation is that TNOs detected in only one or two epochs have scientific value as measurements of the abundance and/or IR colors of small TNOs, even if our knowledge of their orbital state is less complete.

The third motivation for examining all potential two-epoch linkages stems from concerns that the ResNet is overtrained on the PSF of implanted objects, which differs slightly from the on-sky PSF of actual TNOs. Such overtraining could potentially lead the network to misclassify or reject genuine detections of bright TNOs with sufficient S/N to distinguish the two PSFs.

5.2.1. Search among ResNet Rejections

We proceed by collecting the *uv*-plane positions and rates for the ≈4000 single-epoch tracks that pass SMP quality cuts and exhibited S/N > 5, the lower limit placed on successful linkages in Equation (13). We examine all pairs of these good, nonimplanted tracks that occurred in distinct epochs to find any that would fit a bound TNO orbit with distance $d > 6.7$ au. This is accomplished using the model of Equation (10), assuming the line-of-sight motion $\dot{\gamma} = 0$, and scanning across inverse distance γ. The range of γ (if any) that yields a $p > 0.01$ fit to both epochs' positions and rates is recorded.

For each of the ≈5000 two-epoch track pairs meeting this criterion, we next search through all KBMOD detections from the third epoch for any track whose position and velocity are broadly consistent with an orbit passing through the two-epoch pair. This is done *regardless of the ResNet score* for the third track. In this way we should locate any KBMOD track, regardless of ResNet score, that forms a plausible TNO three-epoch triplet with two ResNet-approved, S/N > 5 tracks.[16] If this process is thorough, then given any two of the tracks in one of the 23 known real three-epoch orbits, we should locate the third. Indeed, we rediscover the third epoch in all 69 possible cases.

After removing all of the new triplets that are duplicates of one of the 23 known triplets, we perform single-epoch and three-epoch SMP on each of the 277 remaining triplets. Applying the quality cuts described in previous sections yields five newly discovered three-epoch triplets, each by construction containing one track that was rejected by the ResNet cuts. One of these, at $m_{AB} = 24.5$, is a fourth detection of the

[16] Note that searching the large collection of ResNet-rejected KBMOD detections is feasible because the requirement to link with two ResNet-accepted epochs enormously reduces the available phase space for candidates. It would be infeasible to bypass the ResNet selection for the entire search.

Table 1
Summary of Each Object Found in ⩾2 Epochs of the JWST Data

Name samples[a]	m_{F150W2} (AB mag)	f_{322}/f_{150}	d_{bary} (au)	i	e_{min}	a_{min} (au)	m_{F150W2}, date/single-epoch S/N[b]			
							Epoch 1	Epoch 2	Epoch 3	Duplicate
JPB23	24.06 ± 0.00	0.18 ± 0.00	46.48	2.°07	0.05	44.3	23.80 ± 0.00	24.10 ± 0.00	24.34 ± 0.01	⋯
(= 2015 GK_{56}) C	...	...	...	...	...	...	−4.941/245.4	−0.447/241.8	+5.643/182.2	...
JPB04	24.47 ± 0.00	0.17 ± 0.00	46.20	1.°85	0.06	43.7	24.48 ± 0.01	24.49 ± 0.01	24.49 ± 0.01	24.43 ± 0.01
AC	...	...	...	...	...	...	−5.438/174.9	−0.861/200.0	+5.156/176.7	−5.462/178.2*
JPB22	25.29 ± 0.01	0.31 ± 0.00	38.37	11.°48	0.03	37.1	25.31 ± 0.01	25.28 ± 0.01	25.27 ± 0.01	⋯
A	...	...	...	...	...	...	−5.438/115.1	−0.861/123.2	+5.156/123.6	...
JPB27	25.84 ± 0.01	0.17 ± 0.01	42.49	1.°40	0.02	43.5	25.77 ± 0.01	⋯	25.90 ± 0.01	⋯
BC	...	...	...	...	...	...	−5.438/97.0	...	+5.156/86.4	...
JPB20	25.89 ± 0.01	0.17 ± 0.01	43.21	1.°10	0.03	44.5	25.89 ± 0.01	25.85 ± 0.01	25.90 ± 0.01	25.91 ± 0.01
AC	...	...	...	...	...	...	−4.964/82.0	−0.470/81.9	+5.506/88.0	−0.541/82.8
JPB21	25.98 ± 0.01	0.15 ± 0.01	44.83	1.°59	0.02	44.2	25.97 ± 0.01	25.98 ± 0.01	26.00 ± 0.01	25.99 ± 0.01
AC	...	...	...	...	...	...	−5.365/90.4	−0.518/84.0	+5.530/87.8	−0.541/77.5
JPB19	26.07 ± 0.01	0.36 ± 0.01	41.02	26.°70	0.01	41.6	26.10 ± 0.01	26.07 ± 0.01	26.05 ± 0.01	⋯
A	...	...	...	...	...	...	−5.581/79.5	−0.932/77.6	+5.009/82.9	...
JPB01	26.70 ± 0.01	0.18 ± 0.02	41.07	1.°05	0.07	44.4	26.72 ± 0.02	26.70 ± 0.02	26.69 ± 0.02	⋯
AC	...	...	...	...	...	...	−4.869/50.6	−0.375/51.2*	+5.715/54.2	...
JPB17	26.71 ± 0.01	0.12 ± 0.01	48.83	7.°43	0.07	45.8	26.73 ± 0.02	26.67 ± 0.02	26.72 ± 0.02	⋯
A	...	...	...	...	...	...	−4.893/48.8	−0.399/50.1	+5.691/53.6	...
JPB18	26.76 ± 0.01	0.34 ± 0.02	39.04	23.°87	0.25	52.1	26.71 ± 0.02	26.86 ± 0.02	26.73 ± 0.02	⋯
A	...	...	...	...	...	...	−4.964/53.1	−0.470/45.2	+5.620/48.1	...
JPB05	26.94 ± 0.02	0.13 ± 0.02	45.13	2.°62	0.02	44.3	26.73 ± 0.02	27.15 ± 0.03	26.97 ± 0.03	⋯
AC	...	...	...	...	...	...	−4.869/45.5	−0.375/33.3	+5.715/40.0*	...
JPB16	27.28 ± 0.02	0.13 ± 0.02	42.75	2.°86	0.04	44.3	27.33 ± 0.03	27.24 ± 0.03	27.27 ± 0.03	⋯
AC	...	...	...	...	...	...	−5.557/31.7	−0.979/32.4	+5.034/33.9	...
JPB15	27.34 ± 0.02	0.14 ± 0.03	45.14	1.°03	0.02	44.1	27.35 ± 0.03	27.38 ± 0.04	27.27 ± 0.03	⋯
AC	...	...	...	...	...	...	−5.462/30.9	−0.885/29.2	+5.131/33.3	...
JPB13	27.49 ± 0.02	0.10 ± 0.02	42.80	2.°82	0.00	42.9	27.65 ± 0.05	27.54 ± 0.04	27.43 ± 0.04	27.35 ± 0.04
AC	...	...	...	...	...	...	−4.917/23.7	−0.422/24.3	+5.667/27.8	−0.399/26.2
JPB14	27.54 ± 0.03	0.22 ± 0.03	37.43	4.°49	0.38	60.7	27.48 ± 0.04	27.53 ± 0.04	27.61 ± 0.05	⋯
A	...	...	...	...	...	...	−4.869/24.6	−0.375/23.9	+5.715/23.6	...
JPB12	27.79 ± 0.03	0.18 ± 0.04	43.27	1.°11	0.01	42.8	27.72 ± 0.05	27.92 ± 0.06	27.74 ± 0.05	⋯
AC	...	...	...	...	...	...	−4.964/19.8	−0.470/18.2	+5.620/21.1	...
JPB29	27.84 ± 0.04	0.09 ± 0.07	41.2–46.9	1.°3–1.°1	0.02–0.14	40.4–54.2	27.81 ± 0.06	⋯	27.87 ± 0.06	⋯
BC	...	...	...	...	...	...	−4.823/19.3		+5.762/18.6	...
JPB11	28.04 ± 0.03	0.26 ± 0.04	45.79	16.°32	0.18	38.6	28.03 ± 0.07	27.96 ± 0.06	27.93 ± 0.05	28.28 ± 0.08
A	...	...	...	...	...	...	−4.917/15.5	−0.422/16.4	+5.667/19.6	+5.506/13.7
JPB10	28.15 ± 0.04	0.33 ± 0.06	60.93	19.°80	0.16	52.4	28.06 ± 0.07	28.10 ± 0.07	28.31 ± 0.08	⋯
A	...	...	...	...	...	...	−5.533/15.6	−0.956/15.9	+5.034/13.9	...
JPB30	28.17 ± 0.05	0.37 ± 0.07	42.8–60.2	16.°3–22.°2	0.06–0.15	40.5–70.5	⋯	28.10 ± 0.07	28.24 ± 0.08	⋯
B	...	...	...	...	...	...	...	−0.956/15.0	+5.059/13.7	...
JPB08	28.19 ± 0.04	0.14 ± 0.06	50.69	3.°33	0.14	59.1	28.23 ± 0.07	28.21 ± 0.08	28.12 ± 0.07	⋯
AC	...	...	...	...	...	...	−5.485/14.2	−0.885/14.2	+5.131/14.8	...
JPB09	28.23 ± 0.04	0.28 ± 0.06	28.26	5.°65	0.12	25.2	28.25 ± 0.07	28.19 ± 0.07	28.24 ± 0.07	⋯
A	...	...	...	...	...	...	−5.365/15.1	−0.518/15.5	+5.530/15.4	...
JPB28	28.26 ± 0.05	0.19 ± 0.08	43.4–48.2	3.°6–3.°2	0.03–0.17	42.2–58.3	28.32 ± 0.08	28.19 ± 0.07	⋯	⋯
BC	...	...	...	...	...	...	−4.964/13.8	−0.470/15.2	...	...

Table 1
(Continued)

Name samples[a]	m_{F150W2} (AB mag)	f_{322}/f_{150}	d_{bary} (au)	i	e_{min}	a_{min} (au)	m_{F150W2}, date/single-epoch S/N[b]			
							Epoch 1	Epoch 2	Epoch 3	Duplicate
JPB26	28.29 ± 0.05	0.27 ± 0.06	40.5–48.8	1.°3–1.°2	0.09–0.26	37.0–65.8	28.45 ± 0.10	28.20 ± 0.07	⋯	28.24 ± 0.07
BC	...	...	...	...	...	...	−4.894/11.3	−0.056/15.2	...	−4.775/15.0
JPB06	28.58 ± 0.06	0.29 ± 0.08	42.28	2.°75	0.04	44.0	28.72 ± 0.12	28.43 ± 0.09	28.61 ± 0.10	⋯
AC	...	...	...	...	...	...	−5.509/9.1	−0.932/11.4	+5.083/11.1	...
JPB25	28.60 ± 0.06	0.41 ± 0.11	54.7–279.4	3.°3–164.°4	0.05–0.08	57.6–258.4	⋯	28.73 ± 0.13	28.40 ± 0.09	28.72 ± 0.12
B	...	...	...	...	...	...	...	−0.494/8.6	+5.131/12.2	+5.595/8.7
JPB07	28.60 ± 0.07	0.16 ± 0.08	47.77	1.°76	0.05	45.7	28.56 ± 0.11	28.60 ± 0.12	28.66 ± 0.12	⋯
AC	...	...	...	...	...	...	−5.365/9.6	−0.518/9.4	+5.530/8.9	...
JPB24	28.79 ± 0.07	−0.09 ± 0.11	43.32	1.°10	0.00	43.4	28.65 ± 0.10	29.00 ± 0.17	28.73 ± 0.11	⋯
BC	...	...	...	...	...	...	−5.509/10.2	−0.932/7.4*	+5.083/9.7	...
JPB02	28.97 ± 0.08	0.36 ± 0.12	81.67	4.°96	0.31	62.2	28.98 ± 0.14	28.88 ± 0.13	29.05 ± 0.15	⋯
...	...	...	...	...	...	...	−5.509/7.8*	−0.932/9.0	+5.083/7.1	...
JPB03	29.34 ± 0.12	−0.07 ± 0.16	54.78	24.°54	0.65	33.1	29.30 ± 0.21	29.37 ± 0.22	29.34 ± 0.19	⋯
...	...	...	...	...	...	...	−5.388/5.5	−0.541/5.4	+5.530/5.9*	...

Notes. The distance from the solar system barycenter is d_{bary}; a_{min} and e_{min} are the smallest possible values, i.e., the case of zero radial velocity; the LW/SW flux ratio is defined as unity for equal *AB* magnitudes. Ranges for orbital parameters within 2σ of the maximum likelihood are given when astrometry is available for only two epochs; otherwise the uncertainties are small. Dates of observations are MJD−59974.

[a] Membership in characterized sample A or B is noted by the matching letter. "C" indicates a member of the dynamically cold group.

[b] The S/N is that measured for the single-epoch track, not forced to fit an orbit. An asterisk after the S/N indicates that the track was not discovered through normal procedures and cuts. See Appendix B for explanations.

previously known triplet JPB04. Two others are new three-epoch TNOs, JPB01 and JPB05, at $26.5 < m_{AB} < 27$, where overtraining of the ResNet might be an issue. The detections that the ResNet rejected have S/N > 30, and they are flawless under visual inspection. The ResNet successfully classifies all of the implanted TNOs in this brightness range as real. It is justified, therefore, to include JPB01 and JPB05 in our sample of three-epoch TNOs, and still consider the implants' results as a valid measure of the detection efficiency of the survey, even though the implants were subjected to the ResNet cut but these two failed it—we chalk this up to a difference in ResNet efficiency at high S/N between real and implanted sources.

The other two links of two ResNet-approved tracks to a ResNet-rejected track in the third epoch are JPB02 and JPB03. These are fainter than any ResNet-approved three-epoch TNO, at $m_{AB} \geqslant 29$. In this regime we know that there are many spurious detections and our discovery efficiency is limited by the ability of ResNet to distinguish spurious from real tracks. It is therefore unsurprising that turning off the ResNet filter for one epoch yields fainter linkages—but we should *not* use these in our analyses because (a) they may not be real, and (b) our implant-derived detection efficiency assumes the ResNet filter is in place. We include these two sources in Table 1 and the auxiliary data files as examples of fainter TNOs that *could* be real, but are too faint for us to reliably distinguish from spurious detections or use to infer population statistics.

5.2.2. Follow-up of S/N > 8 "Orphan" Tracks

Returning to the question of whether there are two- or one-epoch TNO detections among our good, nonimplanted tracks, we examine by eye the 111 single-epoch tracks that are not implants, are not apparitions of one of the 21 known three-epoch TNOs, pass all quality cuts, and have S/N > 8 from SMP fitting. At this S/N level, one can readily distinguish real from spurious sources by eye, and Figure 9 shows that this is the S/N level at which there are probably similar numbers of real and spurious tracks passing the single-epoch cuts. For S/N < 8, we do not attempt to distinguish between real and spurious detections that lack confirmation from linking to two other epochs. Figure 9 shows that the number of such "orphan" tracks grows very rapidly as S/N decreases (doubling for S/N < 7), which means that the fraction of real sources is dropping rapidly. We thus do not have confidence that we could exclude false positives from a sample of two-epoch linkages that involve S/N < 8 tracks.

Among the ≈50 orphan S/N > 8 tracks that visual inspection confirms as real, we search for all pairs that can plausibly be fit to a bound orbit of a body at distance > 15 au. These pairs are listed in Table 1, where they are recognizable as missing data from one of the three epochs. With the observed positions from two epochs, we can strongly constrain a linear combination of the TNO's distance and its transverse velocity in the ecliptic plane ($\dot{x}$). The degeneracy between these two quantities is only weakly broken by the rates of motion observed within each of the two detected epochs. The apparent position of the TNO at any given time in our 10-day window is thus constrained to lie within $<1''$ of a line segment on the sky that is parameterized by the TNO's distance d. We thus can determine, for any of our JWST exposures, for what ranges of d (if any) the TNO would have appeared on a detector in that exposure. In each case where the missing third detection could have appeared on a detector, we searched for it again both by eye and by rerunning KBMOD with a lower detection threshold over this limited range of sky position and rate. The results of these searches are detailed in Appendix B. Six of the seven two-epoch TNOs have good geometric reasons for being missed in their third epoch—the orbit could be in a gap or off the edge of the detectors, or, in the case of JPB24, atop a bright galaxy (we recovered this track). The only mysterious case is JPB25, which was imaged in a high-quality region of a detector in epoch 2 for any plausible distance, but further searching for S/N > 4.5 tracks turned up nothing. An extreme–but not impossible–2 × flux dimming during this epoch could explain its disappearance (e.g., contact binary shapes (M. R. Showalter et al. 2021)).

The net result of the search for TNOs is thus as follows:

1. 21 TNOs (including 2015 GK_{56}) and 1 centaur that have reliable detections in all three JWST epochs.
2. 2 TNOs, JPB02 and JPB03, that link three epochs only by bypassing the ResNet in one epoch, and are too faint to be considered reliable.
3. 7 TNOs that were detected at S/N > 8 in two epochs and undetected otherwise.
4. 25 single-epoch detections at S/N > 8 that do not link with detections in another epoch.

Summaries of the available photometric and dynamical information on the 30 two- or three-epoch detections are given in Table 1 and shown in Figure 15, and for the 25 singlets in Table 2. While the singlets are certainly real sources, we do not have sufficient information to deem them to be outer-solar-system bodies. Visual inspection shows that 12 of them (marked as "static" in Table 2) lie atop images of bright galaxies and stars, and all of these have measured rates of motion that are within 3σ of being zero. It thus seems likely that these are all variable stars in the Milky Way or other galaxies. Another (JPB39) has an apparent rate of motion that requires it to be an outer asteroid. The dynamical circumstances of the other 12 sources are indeterminate, but we include their information in Table 2 and in the auxiliary data files for reference.

Figure 16 shows the F150W2 images of TNOs spanning the range of magnitudes found in this survey, from 2015 GK_{56}, previously discovered in ground-based images, at $m = 24.1$, to JPB03, which is >100 × fainter at $m = 29.3$, where we cannot assure freedom from false positives. No objects this faint produced three ResNet-approved tracks passing single-epoch SMP cuts.

5.3. Orbital Characteristics

Detection of a TNO in three distinct epochs over the 10 day JWST observing period yields precise estimates of five of the six degrees of freedom in the TNO's state vector. The apparent angular acceleration of the TNO is the reflex of the Earth's acceleration transverse to the line of sight, and determines the observer–TNO distance to high precision. The transverse angular position and velocity are very well measured as well, and combine with the distance to determine the full Cartesian space position and the two transverse components of its velocity. Any line-of-sight velocity, however, has an effect on the observed positions that is smaller than the systematic errors in establishing the JWST astrometric solutions (see Section 3.3), and is therefore either undetectable or unreliable in our solutions. All of the orbital solutions used in this work

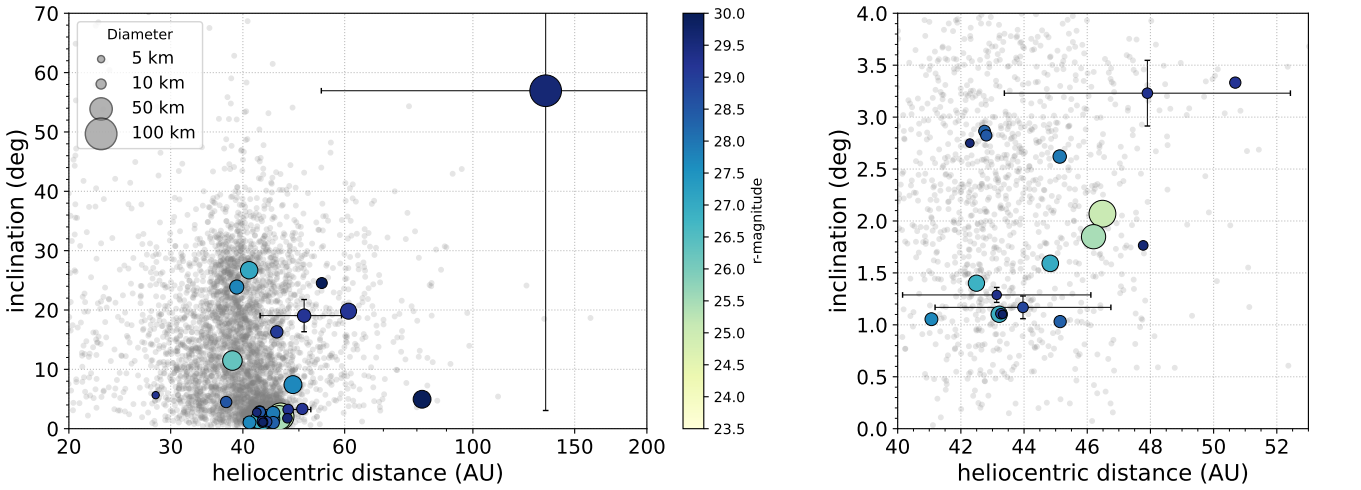


Figure 15. Left: inclination and heliocentric distances of 30 objects detected in three or two epochs of this survey, color coded by their brightness. The marker sizes are scaled to their diameters in kilometers by assuming a 15% albedo for the CCs and 8% for the hot classicals. Heliocentric distances and inclinations are taken from Table 1; for objects with a range of values, the median is plotted, and error bars represent the minimum and maximum of the range. Right: zoomed-in view of the left panel, highlighting the detections in the CC kernel. In both panels, gray circles represent known TNOs from the MPC.

set the line-of-sight velocity to zero. The orbit-level SMP for the three-epoch TNOs yields an estimate of this restricted state vector, and its uncertainties, which are reported in the auxiliary data files. State vectors and derived quantities are given for the nominal epoch JD 2459974.5 (TDB).

Since the line-of-sight velocity is almost the same as the radial velocity from the solar system barycenter, the barycentric distance and angular momentum of a TNO are well determined, but we can only set a lower bound on its total energy. We report in Table 1 the barycentric distance $d_{\rm bary}$ of the TNO at the nominal time and the ecliptic inclination angle i of the TNO, both of which are well constrained. The semimajor axis a and eccentricity e are, however, degenerate because of the unknown line-of-sight velocity $v_{\parallel}$, so we do not report them. We can, however, ascertain the *minimum* possible values of these elements, which occur for $v_{\parallel} = 0$, as

$$e_{\rm min} = \left| 1 - \frac{r v_{\perp}^2}{GM} \right|, \tag{14}$$

$$a_{\rm min} = r\left(2 - \frac{r v_{\perp}^2}{GM} \right)^{-1}, \tag{15}$$

where r is the barycentric radial location, $v_{\perp}$ the nonradial component of velocity, and M the solar system mass. These quantities are reported in Table 1 and the auxiliary data files (Appendix C), and have unimportantly small measurement errors for the three-epoch TNOs.

For the two-epoch TNOs, there is, as noted above, an additional degeneracy among the state vector elements because the weak measure of the angular acceleration leaves the radial distance $d_{\rm bary}$ is weakly constrained. For JPB27, this degeneracy is broken by clear detection of the body on the contemporaneous HST WFC3 images of the field reported in A. Morgan et al. (2026). For the other five two-epoch objects, we can determine $a_{\rm min}$, $e_{\rm min}$, and the expected sky position and rate of the body in any observation by asserting a value of $d_{\rm bary}$. We then scan the range of $d_{\rm bary}$ to determine the following:

1. The $d_{\rm bary,ML}$ value that produces the minimum χ^2 of the difference between the orbit model and the measured positions and rates at the two epochs.
2. The bounding values ($d_{\rm bary,min}$, $d_{\rm bary,max}$) that produced χ^2 values that are 4 times larger than the minimal χ^2 value from step (1). This is a not-rigorous $\pm 2\sigma$ confidence region.
3. We clip ($d_{\rm bary,min}$, $d_{\rm bary,max}$) to the range that yields a bound orbit.
4. We eliminate from the ($d_{\rm bary,min}$, $d_{\rm bary,max}$) interval any values of $d_{\rm bary}$ for which the TNO would have fallen onto a usable part of a NIRCam SW detector in the missing epoch. This part of the inference depends upon an assumption that each of these objects was bright enough in its third epoch to be detectable at near 100% efficiency.

Appendix B reports the constraint methods applicable to each two-epoch TNO. The columns for $d_{\rm bary}$, i, $a_{\rm min}$, and $e_{\rm min}$ in Table 1 for the two-epoch TNOs report the range of values obtained across the inferred possible range of $d_{\rm bary}$. The auxiliary data described in Appendix C contain all of the dynamical properties at $d_{\rm bary,min}$, $d_{\rm bary,ML}$, and $d_{\rm bary,max}$.

An additional detail about the determination of orbital elements; for those TNOs that are detected on two distinct tiles in a given epoch, we make use of only one observation per epoch when determining the state vector and elements. This is because the systematic uncertainty in astrometric registration of different tiles (Section 3.3), combined with the small time interval between the observations and very high precision of the position determinations, can force the orbit to fit a spuriously high rate of motion between these two observations. We obtain a more reliable result by omitting one, preferably retaining a measure that is on the same tile as the other epochs' detections.

5.4. *Long-wave (LW) Data*

Every NIRCam exposure collected data in filter F150W2 on the SW detectors, and in filter F322W2 on the LW detectors. We do not search the LW images for TNOs since the SW images will reach higher S/N for every TNO. No implantation

Table 2
Unlinked Detections

Name	m_{F150W2} (AB mag)	f_{322}/f_{150}	Rate (arcsec/day)	Static?
JPB53	26.56 ± 0.02	0.22 ± 0.03	(+ 3.65, + 6.94) ± (0.13, 0.13)	
JPB47	27.13 ± 0.04	0.45 ± 0.06	(− 0.28, − 0.51) ± (0.23, 0.22)	*
JPB44	27.28 ± 0.06	0.35 ± 0.10	(− 0.45, + 0.24) ± (0.35, 0.35)	*
JPB41	27.43 ± 0.05	0.37 ± 0.07	(− 0.18, − 0.07) ± (0.33, 0.32)	*
JPB49	27.53 ± 0.04	0.56 ± 0.07	(+ 0.40, + 0.29) ± (0.27, 0.26)	*
JPB52	27.68 ± 0.06	0.59 ± 0.12	(+ 0.37, + 0.34) ± (0.37, 0.37)	*
JPB40	27.87 ± 0.06	⋯	(+ 7.93, − 2.37) ± (0.37, 0.34)	
JPB42	27.89 ± 0.06	1.33 ± 0.12	(+ 0.25, + 0.25) ± (0.35, 0.36)	*
JPB45	28.29 ± 0.10	0.80 ± 0.20	(− 1.16, − 0.02) ± (0.62, 0.59)	*
JPB34	28.31 ± 0.13	0.43 ± 0.15	(− 1.60, − 1.05) ± (0.68, 0.67)	*
JPB32	28.35 ± 0.11	0.27 ± 0.20	(− 1.39, − 1.26) ± (0.62, 0.62)	*
JPB39	28.37 ± 0.09	0.35 ± 0.12	(+ 37.13, − 14.63) ± (0.66, 0.64)	
JPB31	28.37 ± 0.10	0.46 ± 0.15	(− 1.28, − 0.11) ± (0.51, 0.51)	*
JPB48	28.43 ± 0.09	0.39 ± 0.12	(− 0.20, − 1.39) ± (0.56, 0.56)	
JPB43	28.52 ± 0.12	0.44 ± 0.20	(+ 0.45, + 0.02) ± (0.75, 0.74)	*
JPB55	28.58 ± 0.11	−0.19 ± 0.15	(− 2.36, + 2.93) ± (0.71, 0.70)	
JPB46	28.60 ± 0.12	0.26 ± 0.14	(+ 4.05, − 7.44) ± (0.76, 0.78)	
JPB37	28.63 ± 0.11	0.20 ± 0.15	(+ 4.87, + 6.75) ± (0.74, 0.73)	
JPB35	28.65 ± 0.11	⋯	(+ 5.79, − 1.52) ± (0.74, 0.66)	
JPB36	28.70 ± 0.11	0.15 ± 0.16	(− 7.90, − 1.13) ± (0.61, 0.59)	
JPB51	28.73 ± 0.13	1.13 ± 0.34	(− 1.41, − 0.28) ± (0.77, 0.76)	*
JPB54	28.77 ± 0.12	0.02 ± 0.17	(+ 4.25, − 4.38) ± (0.77, 0.75)	
JPB50	28.86 ± 0.13	−0.10 ± 0.17	(+ 1.82, − 6.49) ± (0.88, 0.86)	
JPB38	28.88 ± 0.13	0.39 ± 0.18	(+ 0.63, + 1.60) ± (0.96, 0.96)	
JPB33	28.93 ± 0.13	0.30 ± 0.22	(+ 2.07, − 0.51) ± (0.77, 0.79)	

Note. Single-epoch detections that are not plausible links to any other detections. The "Static?" column is marked if the object appears atop a static source (galaxy or star), indicative of it being a variable star rather than a solar system body.

of synthetic sources is done for the LW data since we do not need to characterize discovery efficiencies. We can, however, measure LW fluxes for all objects found in the SW data. We do this in a "forced photometry" mode of the SMP algorithms, whereby the celestial coordinates of the moving TNO are taken from the orbit fit to the SW data, and only the LW fluxes are left as free parameters. Thus, a full F322W2 lightcurve is obtained for every detected TNO, albeit at a lower S/N, which leaves many of the objects without significant flux detections even when stacking all exposures together. Rather than assigning magnitudes to these low-S/N fluxes, we present in Tables 1 and 2 the ratio of each object's F322W2 flux to its flux in F150W2, in units such that a ratio of 1 implies equal AB magnitudes in each band.

Processed data tables giving SW and LW pixel data and derived orbital and photometric properties for all of the objects listed in Tables 1 and 2 are made publicly available at doi: 10.11570/26.0008.

5.5. Characterized Samples

To constrain the magnitude or SDs of TNOs, we need to define selection criteria for observed TNOs such that (1) we are confident that our detected source list is free of false positives, and (2) we can derive, from our implanted sources, the detection efficiency of the survey for the given sample. We create two disjoint samples that should satisfy these conditions.

Sample A is defined as those TNOs detected in three epochs by KBMOD, passing ResNet cuts, passing the single-epoch SMP cuts of Section 4.3.2, and passing full-orbit SMP cuts of Equation (13). In other words, those objects found and linked through the "normal procedures" described in Sections 4 and 5.1, where we demonstrate they should be free of false positives. This includes 18 TNOs, after we exclude the targeted 2015 GK_{56}. To these sources, we add JPB01 and JPB05, because they failed the "normal procedure" only because of a likely overtraining bias of the ResNet against high-S/N real sources. Sample A thus contains 20 three-epoch detections.

Sample B is defined as those sources with NIRCam detections in only two of the three JWST observing epochs, both with S/N > 8 in the SMP analyses of their single-epoch tracks. The absence of false positives in this sample is assured because of the relative ease of distinguishing real from spurious sources by eye at the higher S/N threshold, and also because there are few enough of these high-S/N single-epoch detections (only ≈50; see Figure 9) that the chances of two of them spuriously being linked into a common TNO orbit are small. We detect seven TNOs meeting the sample B conditions.

The reasons that a TNO might be missed in one epoch and end up in sample B instead of A would be:

1. *Geometry.* The object is simply not within the FOV of any images from the missing epoch. This is definitely the case for two of the sample B members, and plausible for three others.
2. *Hiding*. The object was lost to poor subtraction of a brighter overlapping source. This was the case for JPB24, which we did later locate in the third epoch, but we leave it in sample B.

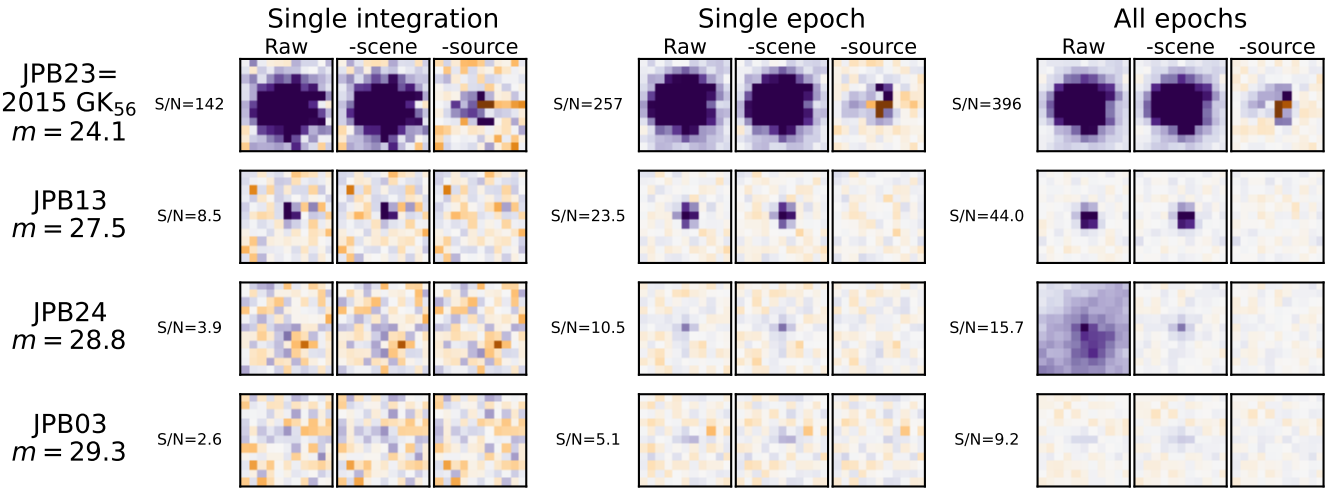


Figure 16. Each row shows representative images of TNOs from the F150W2 search images. Each triplet of cutouts has the original image, the image with background model subtracted, and the image with the TNO model also subtracted. The groups, from left to right, are: images from a single 120 s integration; from a stack of six such integrations in the first observing epoch; and from a stack of all three epochs' images. The top row is 2015 GK_{56}, which was intentionally targeted and is the brightest moving object in the field, and shows significant model residuals due to the finite accuracy of our PSF model. The second row is JPB13, which is 23 × fainter and still confidently detected even in single integrations. Third is JPB24, our faintest confidently detected source, 3 × fainter than JPB13, which requires stacking over an epoch to be clearly visible. The bottom row is JPB03, another 1.6 × fainter, which failed the ResNet filter for one epoch and is in a brightness regime where our discrimination between real and spurious sources is too weak to estimate the real population accurately.

3. *Light curve*. The object faded below the S/N > 5 threshold applied to each epoch for sample A during one of the epochs. This may have been the case for JPB25.

The search efficiency, or equivalently the effective search solid angle, for both samples can be derived from the implant results. For an implanted or real TNO to be detected in some epoch through the normal procedures, two events must occur; first, the TNO must fall onto one of the NIRCam SW detectors. Second, if on the image, the source must be detected by KBMOD, be approved by the ResNet, and yield acceptable SMP S/N. We can use the implants to determine the probabilities of these two events separately.

The geometric probability of a TNO landing on an imaged region in two or three different epochs can be determined numerically simply by counting the number of implants that do so. If we assume that the orbital distribution is independent of the flux distribution for small TNOs, we can use *all* of the implants, regardless of whether they are detected, to calculate the effective area on the sky of the survey for two- and three-epoch capture of TNOs by NIRCam. Doing so for the implants at distances $35 < d_{\rm bary} < 55$ au and inclinations $i < 5°$ yields effective areas of $A_1 = 29$, $A_2 = 56$, and $A_3 = 139$ arcmin^2 for being imaged onto a NIRCam SW detector in one, two, and three epochs, respectively. (We do not count apparitions within 15 pixels of the edge of the detector.) For the "hot" implants with $i > 5°$ in the same distance range, the values are 60, 81, and 128 arcmin^2—a slightly larger chance of being imaged at all because their apparent motion is larger over the 10 day survey, but also a larger chance of appearing in a gap or falling off the array in at least one epoch.

In Section 4.3.3, we derived a probability $p_{\rm det}(m)$ of any apparition of a TNO of magnitude m creating a track that passes quality cuts, and a variant we will call $p_8(m)$ of this track having S/N > 8. Under the validated assumption of independent detection probability in each epoch, and ignoring light-curve variations of a given TNO, the effective area of sample A at magnitude m becomes $A_3 p_{\rm det}^3(m)$.

The effective area of sample B is a bit more complex, as a TNO can enter the sample in two ways. First, it can appear in only two epochs and be detected at S/N > 8 in both, yielding an effective area of $A_2 p_8^2(m)$. Alternatively, it can appear on the NIRCam array in three epochs, achieve S/N > 8 on two of them, but fail to be detected in the third epoch (else it would be in sample A). The effective area for this detection channel is hence $3A_3 p_8^2(m)[1 - p_{\rm det}(m)]$.

At magnitudes $m < 28$ where $p_{\rm det}{=}p_8 = 0.96$, the above formulae imply the expected ratio of effective areas for sample B to sample A (and hence the expected ratio of number of detections) would be $\langle N_B/N_A\rangle = 3(1 - p_8) + A_2/p_8 A_3 = 0.54$ and 0.78 for the cold and hot objects, respectively. Our search yielded $N_A/N_B = 5/12$ and $2/8$ for these two dynamical groups, both within the 91% probability region of expected count ratios under Poisson statistics. The effective areas suggest that the number of single-epoch TNO detections at $m < 28$ should be <30% of the number of $m < 28$ TNOs in samples A+B, and a larger fraction at $m > 28$, once we exclude the likely static variables from the singlet list. This is roughly what we find.

6. Binary Detections

The search presented above did not reveal any evidence of binarity in the detected objects. This may be the result of a bias to the preferential detection of singletons from our pipeline. For example, the ResNet training set contained only singleton implants, and so it may be possible that the ResNet preferentially classed partially resolved binary sources as false.

To determine if this were the case, the ResNet was retrained using a training set with both binary and single TNO implants. Using a dataset containing binary implants in place of every singleton injected into the original search, we rerun the full calibration process (Section 3). To avoid injection into the identical pixels, each artificial binary was offset by 0.″05–0.″32 (≈1.2–10.3 SW pixels) from the location of the singleton implant. The primary binary sources were implanted with the same brightness as used for the singletons. The secondary

components were assigned random brightness and spatial offsets, drawn uniformly from 1% − 100% of the primary's brightness and 0.″05–0.″20 (≈1.2–6.5 SW pixels), respectively. After the full calibration, subtraction, and shift-and-stack steps, the ResNet was retrained as discussed in Section 4.2, but using detected implanted sources from both the singletons of the original search and the binary implants.

After classification using this new ResNet model, all candidate nonimplant sources were visually inspected. This new search only revealed the exact set of detections as found in the original search, and did not detect any binary sources.

An experiment was performed to assess the significance of a null binary set. In particular, we asked the question; if objects in this survey shared the same single-binary ratio and binary properties as the Solar System Origins Legacy Survey (SSOLS; S. B. Porter et al. 2024) dataset, how many binaries should we have detected? To that end, we assume that binaries from this survey would have similar primary–secondary brightness ratios and separation distributions (Δr, in units of the system's Hill sphere) as the SSOLS dataset. We also assume that any hidden binaries share identical densities with their SSOLS counterparts. Under these assumptions, the apparent primary–secondary separation $\Delta r'$ of a hypothetical binary in our survey with brightness m_{F150W2} is given by

$$\Delta r' = \Delta r \left(\frac{m'}{m} \right)^{2/3}, \tag{16}$$

where Δr is the apparent primary–secondary separation of the SSOLS binary, m and m' are the masses of the primary bodies from the SSOLS and this survey, respectively. Both m and m' are inferred directly from their apparent brightnesses assuming (F606W − F150W2) = 0.8.

For each object i detected in our survey, we determined $\Delta r'_{ij}$ from Equation (16) for each binary j, observed by SSOLS. Then, using the observed primary–secondary brightness ratio Δm_j, we determined the probability P_{ij} of detection of such a system from the outcomes of the ResNet trained on the binary +singleton sample.

Additionally, we also performed a simple numerical experiment to evaluate the consistency with the SSOLS dataset. For each detection i in this survey, a random number was drawn; if it was less than or equal to the SSOLS observed binary fraction (24 resolved binaries in 204 targets), it was treated as a binary. For these binary detections, three additional random numbers were drawn to simulate three observation epochs, and a binary was flagged as detected if any of these numbers were less than the detection probability P_{ij}. We repeated this procedure across all survey detections for 5000 trials to determine if, in one iteration of the experiment, a binary was detected in at least one of three epochs. In only 9.5% of cases was a binary TNO numerically detected, meaning that no detections would be expected in the vast majority of cases. We thus conclude that the lack of binary TNO detections in this survey is fully consistent with scaled expectations from the SSOLS dataset and should not be interpreted as a change in binary properties of the fainter sources of this survey.

We highlight that the majority of weight in the above calculation is held by the three brightest detections of our survey, with $P_{ij} = 0$ for all other targets. At such small sizes, only very widely separated binaries (with respect to their Hill spheres; A. H. Parker et al. 2011) would be detected.

7. The Luminosity Function (LF)

The goal of this section is to identify the functional forms of the differential surface density that are most consistent with the survey data. Given an assumed distribution $\Sigma(m)$ and a survey's areal coverage Ω, the expected number of detections is given by:

$$\bar{N}(\theta) = \Omega \int \eta(m)\, \Sigma(m, \theta)\, dm, \tag{17}$$

where the detection efficiency of the survey, $\eta(m)$, describes the probability of successfully detecting an object with mean magnitude m that lies within the survey's geometric area, Ω. The η function is the detection probability described in Section 4.3.3. The vector θ holds any free parameters of the Σ function.

Given some assumed functional representation, the shape of the LF can be quantified by applying a maximum-likelihood fit to the observed data. With this approach, we obtain parameter values that best describe the observed distribution of magnitudes in the data. So for a single survey with detections indexed by i, we have

$$\mathcal{L}(\theta) \propto e^{-\bar{N}(\theta)} \prod_{i=1}^{N} \Sigma(\hat{m}_i, \theta)\, \eta(\hat{m}_i). \tag{18}$$

We note that the assumptions leading to Equation (18) differ somewhat from those adopted in previous studies (B. Gladman et al. 1998, 2001; J.-M. Petit et al. 2006; W. C. Fraser & J. Kavelaars 2008; W. C. Fraser & J. J. Kavelaars 2009; W. C. Fraser et al. 2010, 2014; K. J. Napier et al. 2024), which were all based from a statistical approach presented in T. J. Loredo (2004). Our final likelihood form was derived based on two assumptions. First, the measured magnitude $\hat{m}$ of each source is a good approximation of its true magnitude m, and second, the detection of a source does not necessarily result in a real and valid measurement of its brightness (i.e., the search process is nondeterministic). While for the observations presented here, we were able to measure a valid brightness for every detection, one can realize certain circumstances where this is not always true. A detailed derivation of this likelihood function is presented in Appendix D.

Despite our assumptions and the likelihood form that results from those assumptions, we tested all variations of the above assumptions. That is, we also fit using the form by T. J. Loredo (2004) that omits the efficiency function in the second term, and we also tested both variations when assuming the observed likelihood is treated as a Gaussian variable. In all cases, the resultant fits were nearly identical. For example, the full A sample resulted in best-fit alphas of $\alpha = 0.27^{+0.07}_{-0.06}$, $0.26^{+0.07}_{-0.06}$, $0.26^{+0.07}_{-0.06}$, $0.27^{+0.07}_{-0.06}$ for the likelihood forms A ($\hat{m} \approx m$, with $\eta(m)$ factor), B ($\hat{m} \sim$ Gaussian variable, with $\eta(m)$ factor), C ($\hat{m} \approx m$, no $\eta(m)$ factor), and D ($\hat{m} \sim$ Gaussian variable, no $\eta(m)$ factor), respectively. We conclude that for this project, the adopted likelihood form makes little difference.

7.1. Sample Selection

As mentioned in Section 5.5, we created two disjoint detection samples to ensure the catalog is free of false positives and to measure the survey's detection efficiency using implanted sources. For the following LF analysis, we

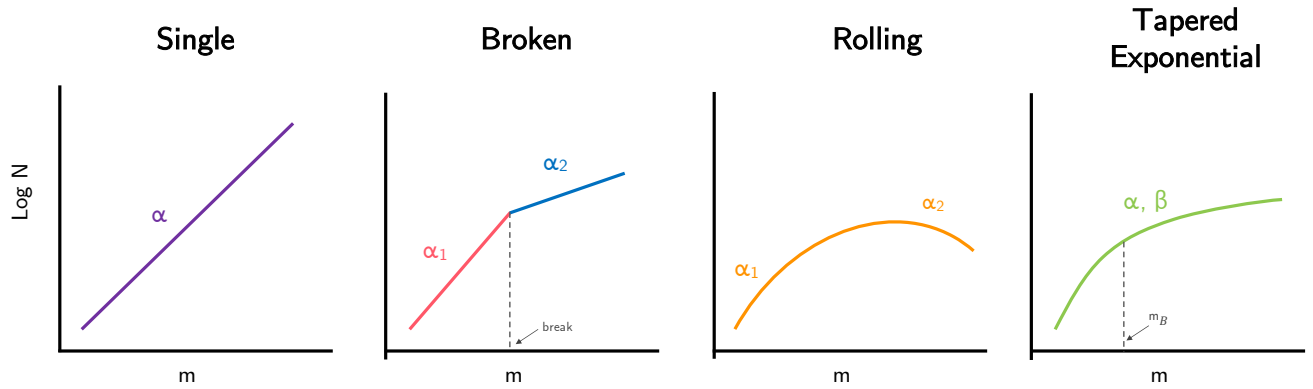


Figure 17. A schematic showing the four functional forms that describe the TNO SD. Each form describes the distribution of objects across different magnitudes m and is parameterized by specific slopes and transition points that reflect changes in object number density. Note that while our detections lie beyond the magnitude break, we include the broken power-law form in this schematic simply for visual comparison.

perform two separate analyses using these datasets: one based on all three-epoch detections (sample A), and another based on the combined set of three-epoch and two-epoch detections (sample A+B).

We fit several functional forms of varying complexity to model our datasets (see Figure 17), and use Markov Chain Monte Carlo (MCMC) sampling to estimate parameter uncertainties and identify the best-fit solutions. For analysis, we define three dynamical groups in our survey:

1. The *full* sample includes all objects detected at barycentric distances $d > 30$ au. This cut excludes one centaur from our total detections, as its dynamical evolution likely differs from that of the current TNO population (R. Brasser et al. 2012; K. Volk & R. Malhotra 2013).
2. The *cold* population sample is a subset of the full object sample, defined by barycentric distances $38 < d < 55$ au and ecliptic inclinations $i \leqslant 5°$.
3. The *hot* population is a complement of the cold sample in the full sample. These are objects with $d > 55$ au or within any distance range beyond Neptune but with inclinations $i > 5°$.

Our separation of the hot and CC components at $i = 5°$ is not definitive. It is the unforced component of the ecliptic inclination (i.e., free inclination) that shows a clear delineation between the dynamically cold, largely undisturbed population and the dynamically excited population of the Kuiper Belt (C. Van Laerhoven et al. 2019; Y. Huang et al. 2022). While we do not have free inclinations for our detections, the hot and cold components share comparable apparent magnitude distributions, so a few misclassifications would not significantly change our results.

For the purposes of the $\Sigma(m)$ analysis, we transform our objects' F150W2 filter magnitudes to the conventional r-band magnitude. To estimate the color difference between the F150W2 filter (effective wavelength $\sim$1.7 μm) and the r filter (effective wavelength $\sim$0.6 μm), we use the reflectance spectrum of the CC TNO Arrokoth (W. Grundy et al. 2020; D. P. Cruikshank et al. 2020). We also calculate the $r - $ F150W2 color of the Sun from values given in C. N. A. Willmer (2018). Taking all of these together, we adopt $r - $ F150W2 ≈ 1.0.

We point out that there remain possible systematic issues with inter-instrument calibrations between LEISA and MVIC that were utilized for the Arrokoth spectrum in Figure 2 (C. J. A. Howett et al. 2017; S. Protopapa et al. 2020; M. Fayolle et al. 2021). Such differences would directly manifest in the $r - $ F150W2 estimate we present, and some appropriate caution is warranted.

7.2. Single Power-law Fits

The LF of the Kuiper Belt as a whole, as well as its dynamically distinct cold and hot classical components, has been extensively studied over the past two decades (B. Gladman et al. 2001; G. M. Bernstein et al. 2004; W. C. Fraser & J. Kavelaars 2008; W. C. Fraser et al. 2008; C. I. Fuentes et al. 2009; K. J. Napier et al. 2024). We build on the methodologies established in these previous works by applying functional forms of varying complexity to model our dataset and assess how well each captures the observed distribution. We begin by fitting a single power-law model, described by

$$\Sigma_{\rm single}(m) = 10^{\alpha(m-m_0)}, \tag{19}$$

where m_0 is the magnitude at which the surface density equals one object per square degree, and α is the power-law slope.

Fitting this model to samples A and A+B yields best-fit slopes of $\alpha = 0.27^{+0.06}_{-0.07}$ and $\alpha = 0.29^{+0.06}_{-0.07}$, respectively. For sample A, we find $\alpha = 0.27^{+0.08}_{-0.09}$ for the cold population and $\alpha = 0.24^{+0.09}_{-0.10}$ for the hot population, with similarly consistent values obtained for sample A+B. The agreement within uncertainties across all subsamples indicates that the faint-end slope of the LF does not differ significantly between dynamical groups (see Table 3).

It is worth noting that earlier studies have already established that the LF breaks or transitions from a steep slope, for bright and large objects, to a shallower distribution for fainter and smaller objects. The transition or break magnitude occurs at $m_r \sim 25$ (G. M. Bernstein et al. 2004; C. I. Fuentes et al. 2009; K. J. Napier et al. 2024). Our data consist exclusively of objects fainter than this break and therefore probe only the fainter end of the LF. As a result, the observations do not provide meaningful constraints on the bright-end behavior or the location of the break, and we therefore do not attempt to fit a broken power-law model. Instead, a single power law is sufficient to characterize the

Table 3
Best-fit Differential Surface Density Models

Sample	Single Power Law		Rolling Power Law				Exponentially Tapered Power Law				
	α	m_0	α_1	α_2	Σ_{23}	ΔBIC	α	β	m_0	m_B	ΔBIC
Hot A	$0.24^{+0.10}_{-0.09}$	$21.53^{+2.10}_{-3.75}$	$0.86^{+0.09}_{-0.09}$	$-0.09^{+0.02}_{-0.02}$	$0.37^{+0.10}_{-0.10}$	0.9	$0.18^{+0.06}_{-0.05}$	$0.14^{+0.05}_{-0.05}$	$16.29^{+3.85}_{-6.20}$	$25.23^{+3.50}_{-6.25}$	3.8
Cold A	$0.27^{+0.09}_{-0.08}$	$21.38^{+1.79}_{-2.64}$	$0.92^{+0.14}_{-0.14}$	$-0.08^{+0.02}_{-0.02}$	$0.19^{+0.10}_{-0.09}$	1.9	$0.20+^{+0.07}_{-0.06}$	$0.18^{+0.10}_{-0.10}$	$17.03^{+3.37}_{-5.58}$	$25.10^{+3.14}_{-6.25}$	4.6
Full A	$0.27^{+0.07}_{-0.06}$	$20.56^{+1.53}_{-2.10}$	$0.87^{+0.08}_{-0.08}$	$-0.07^{+0.01}_{-0.01}$	$0.39^{+0.09}_{-0.09}$	0.2	$0.21^{+0.07}_{-0.05}$	$0.14^{+0.05}_{-0.05}$	$16.07^{+3.61}_{-5.92}$	$25.63^{+3.10}_{-6.19}$	3.9
Hot AB	$0.28^{+0.11}_{-0.09}$	$22.46^{+1.73}_{-2.58}$	$0.83^{+0.08}_{-0.08}$	$-0.08^{+0.02}_{-0.02}$	$0.35^{+0.11}_{-0.10}$	2.9	$0.19^{+0.07}_{-0.06}$	$0.13^{+0.05}_{-0.05}$	$16.17^{+3.61}_{-6.36}$	$26.37^{+3.61}_{-6.65}$	3.8
Cold AB	$0.29^{+0.08}_{-0.07}$	$21.61^{+1.51}_{-2.20}$	$0.88^{+0.13}_{-0.13}$	$-0.07^{+0.02}_{-0.02}$	$0.19^{+0.09}_{-0.08}$	2.8	$0.21^{+0.06}_{-0.06}$	$0.17^{+0.10}_{-0.10}$	$17.11^{+3.02}_{-5.52}$	$25.89^{+2.99}_{-6.25}$	6.1
Full AB	$0.29^{+0.07}_{-0.06}$	$20.89^{+1.43}_{-1.74}$	$0.85^{+0.08}_{-0.07}$	$-0.07^{+0.01}_{-0.01}$	$0.38^{+0.09}_{-0.10}$	3.5	$0.22^{+0.07}_{-0.06}$	$0.14^{+0.05}_{-0.05}$	$16.43^{+3.38}_{-6.06}$	$26.34^{+2.93}_{-5.69}$	6.8

Note. $\Delta\text{BIC} = \text{BIC}_m - \text{BIC}^*$, where BIC^* is the reference model (here, the single power law) and BIC_m is any candidate model. The size of ΔBIC shows how strongly the candidate model is disfavored compared to the single power-law model.

Table 4
Informative Priors

Sample	Rolling Power Law			Exponentially Tapered Power Law			
	α_1	α_2	Σ_{23}	α	β	m_0	m_B
Hot	0.9 ± 0.1	-0.1 ± 0.3	0.4 ± 0.1	0.2 ± 1.0	0.1 ± 0.5	14.0 ± 11.4	29.1 ± 27.0
Cold	1.0 ± 0.2	-0.1 ± 0.4	0.2 ± 0.1	0.2 ± 1.0	0.2 ± 1.0	12.6 ± 8.0	28.4 ± 25.0
Full	0.9 ± 0.1	-0.1 ± 0.3	0.4 ± 0.1	0.2 ± 1.0	0.1 ± 0.5	14.3 ± 11.4	29.1 ± 27.0

Note. We adopt the best-fit values from K. J. Napier et al. (2024) as informative priors for our rolling and exponentially tapered power-law fits. Since the distributions in their analysis were approximately Gaussian, we employ Gaussian priors centered on these reported values. For parameters governing the bright end of the LF and normalization values (α_1, Σ_{23}, m_0), we use the reported uncertainties. Alternatively, for parameters influencing the faint end (α_2, α, β, m_B) of the function, where most of our data lies, we broaden the Gaussian widths by 10σ to allow for greater exploration of the parameter space.

faint-end slope sampled by the magnitude range of our observations.

7.3. Rolling Power-law Fits

As another level of complication, we fit a rolling power-law model by Equation (18). Unlike the broken power law, which introduces a sharp transition between the bright-end and faint-end slopes (α_1 and α_2), the rolling power law allows for a gradual change in slope across magnitudes. This smoother transition is achieved by including a quadratic term in the exponent, allowing the logarithmic slope to change gradually with magnitude instead of shifting suddenly, and is given by

$$\Sigma_{\text{rolling}}(m) = \Sigma_{23} 10^{\alpha_1(m-23)+\alpha_2(m-23)^2}, \quad (20)$$

where Σ_{23} is the number of objects with $m_r = 23$ per square degree, while α_1 and α_2 are the bright- and faint-end slopes, respectively, which control the shape of the function.

To account for the fact that our dataset provides no leverage on the LF brightward of $m_r \sim 25$, we adopt the fitting results from K. J. Napier et al. (2024) as informative priors for our MCMC sampling. We adopted Gaussian priors centered on their reported best-fit values because the distributions in those fits were close to being Gaussian (see Table 4). This approach incorporates existing knowledge and effectively constrains the parameter space explored during the sampling process and is applied to both the current and subsequent functional fits.

7.4. Exponentially Tapered Power-law Fits

Finally, we fit an exponentially tapered power law to our differential distribution. This functional form is seen in numerical simulations of planet formation by SI+PCC (U. Schäfer et al. 2017; R. Li et al. 2019) and is described by

$$\Sigma_{\text{tapered}} = \ln(10) \cdot \exp\left(-10^{\beta(m_B-m)}\right) \cdot 10^{(\alpha-\beta)m-\alpha m_0} \cdot \left(\alpha \cdot 10^{\beta m} + \beta \cdot 10^{\beta m_B}\right). \quad (21)$$

Here, α is the faint-end slope of the distribution, β is the strength of the exponential bright-end behavior, m_0 is a normalization factor, and m_B is the magnitude where the exponential term starts to overpower the power-law term. We again use the K. J. Napier et al. (2024) results as Gaussian priors, but taking into consideration the phase curve effects at m_0 and m_B parameters, as our JWST observations were done near quadrature,[17] rather than near opposition where their fit was derived. We plot our fit results in Figures 18 and 19.

7.5. Comparison of Fits

To determine which Σ function is a better representation of our data, we make use of the Bayesian information criterion (BIC; R. E. Kass & A. E. Raftery 1995),

$$BIC = k\ ln(n) - 2\ ln\ (\mathcal{L}). \quad (22)$$

Here, k denotes the number of parameters in each fit ($k = 2$ for a power law, $k = 3$ for rolling power law, and $k = 4$ for tapered power law), n refers to the sample size, and $ln\ \ (\mathcal{L})$ is the natural logarithm of the likelihood function defined in Equation (18), evaluated at the best-fit (maximum likelihood) parameter values. $ln\ \ (\mathcal{L})$ reflects how well the model fits the data under those parameters, and can thus be interpreted as a goodness-of-fit measure.

[17] In this observing geometry, the larger phase angle reduces the apparent brightness of the object by ~0.35 mag relative to typical near-opposition phase angles due to smaller illuminated area and enhanced surface scattering (see Figure 6 in A. J. Verbiscer et al. 2019).

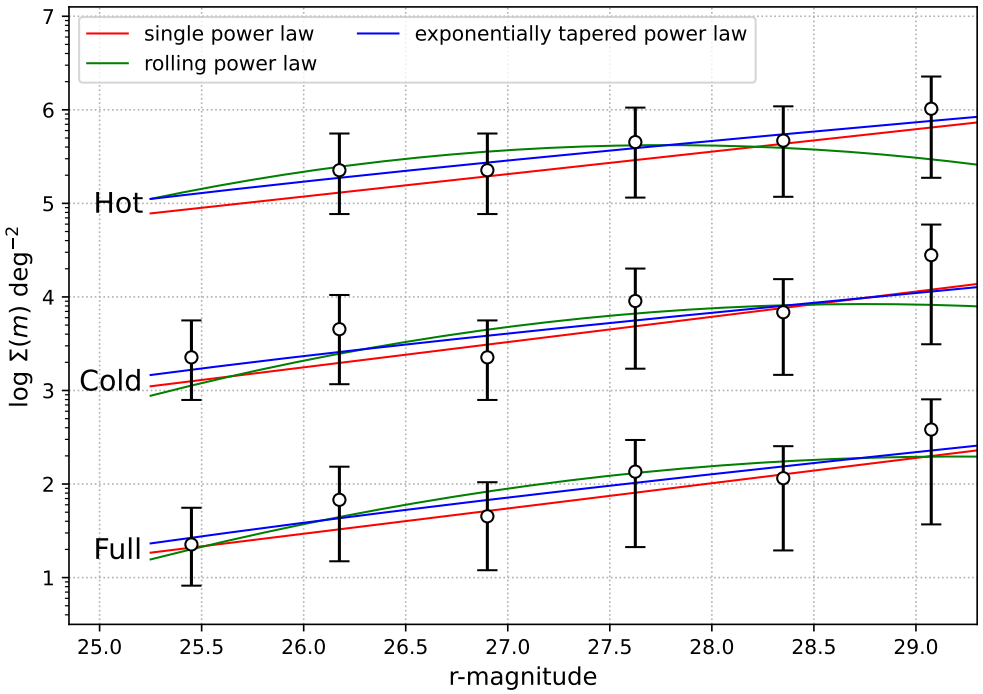


Figure 18. The best-fit differential distributions for the hot, cold, and full TNO of *sample A*, with red, green, and blue lines corresponding to the single, rolling, and exponentially tapered power-law models, respectively. The magnitude distributions are presented in the r band, assuming a nominal color $r - \mathrm{F150W2} = 0.99$ mag. For visual clarity, the cold and hot samples have been vertically offset by 2 and 4 units, respectively. Error bars are the 2σ Poisson uncertainties on the number of objects in each magnitude bin. The best-fitting parameters for each function are given in Table 3.

The BIC imposes a penalty on models with more parameters to discourage overfitting and thereby balance the goodness of fit with the complexity of the model. In general, the model with the lowest BIC value is preferred. To facilitate model comparisons, we consider the difference $\Delta\mathrm{BIC} = \mathrm{BIC}_m - \mathrm{BIC}^*$ which directly measures the relative support for one model over another. Our reference model (i.e., single power law) is denoted by BIC^*, while BIC_m is the candidate model. The value of ΔBIC indicates the strength of evidence against a candidate model being optimal. As a rule of thumb, if ΔBIC is less than 2, the difference is negligible. Values between 2 and 6 indicate positive evidence against the candidate model, while values between 6 and 10 suggest strong evidence (F. J. Fabozzi et al. 2014). When ΔBIC exceeds 10, the evidence against the candidate model is considered very strong. We demonstrate our results in Figures 18 and 19, and summarize them in Table 3.

The single, rolling, and exponentially tapered power laws all fit our data reasonably well. However, based on the ΔBIC values, the single power law is preferred over the more complex rolling and exponentially tapered forms; within our data range, a single power law is sufficient. Adding extra parameters in the more complex models does not yield a significant improvement in fit quality when incorporating the prior from K. J. Napier et al. (2024).

7.6. Color Estimates

One so far unutilized measurement afforded by the NIRCam observations is the LW to SW flux ratio (f_{322}/f_{150}) listed in Table 1. These fluxes can be estimated for objects observed by JWST NIRSpec (N. Pinilla-Alonso et al. 2025), allowing inference of surface type for each of our detections. We adopt the broad three-class scheme: organics, CO_2, or H_2O types (B. J. Holler et al. 2025).

For each class, we adopt the corresponding average NIRSpec reflectance spectrum and derive its behavior in the r band by fitting a linear continuum across 0.71–0.90 μm. This continuum is then extended to shorter wavelengths to approximate the unobserved portion of the spectrum, from which an average (r − F150W2) color is obtained for each compositional category (see Table 5).

Since most orbital solutions derived from our two-epoch detections (sample B) remain insufficiently constrained, we limit our compositional estimates only to the three-epoch detections (sample A). Each sample A object is assigned an (r − F150W2) color based on which spectral type closest matches the object's observed f_{322}/f_{150} flux ratio. This provided r-band brightness estimates for all objects. We find that the CCs are consistent with being drawn from the organics types, and the excited objects are drawn from a mix of classes, consistent with findings from N. Pinilla-Alonso et al. (2025). A companion paper, A. Morgan et al. (2026), discusses the implications of these color measurements. The r-band LF was fit using those values. Best-fit values for each functional form are shown in Table 6, while the assigned classification for each object is listed in Appendix E. We find that the best-fit slopes remain consistent regardless of the adopted colors for each detected object.

8. The Absolute Magnitude Distribution of the CCs

SI+PCC is the currently favored pathway to planet formation because it facilitates the rapid formation of larger planetesimals by overcoming growth barriers (i.e., collisional bouncing, fragmentation, inward drift) (A. Johansen & H. Klahr 2011; J. E. Robinson et al. 2020; D. Nesvorný et al. 2021). Multiple independent studies have investigated the initial mass function of planetesimals formed through SI

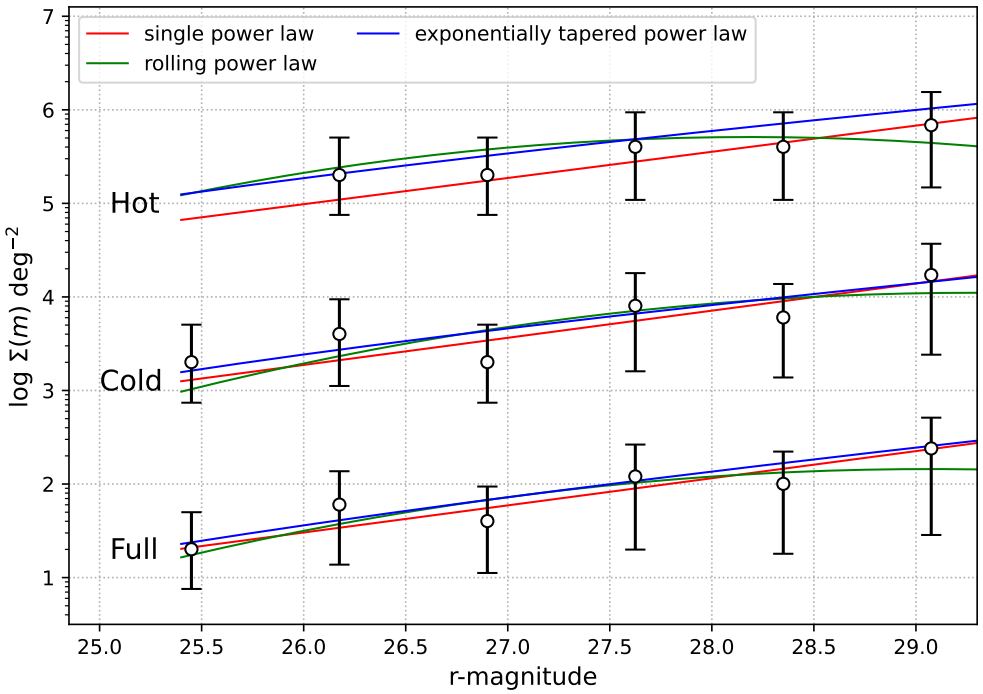


Figure 19. Same as the previous figure but using *sample A+B*. The best-fitting parameters for each function are given in Table 3.

Table 5
Compositional Types with *r*-F150W2 Values and Uncertainties

Object Type	$r-$ F150W2	σ
Cliff organics type	0.61	0.05
DoubleDip CO_2 type	0.34	0.05
Bowl H_2O type	0.30	0.05

Note. The color uncertainty σ was derived from the range of slopes that changed the F150W2 magnitude by no more than the largest photometric uncertainty in our sample ($\pm$ 0.07 mag).

+PCC, but the lower-mass end of this distribution remains poorly constrained due to the limited mass resolution in simulations (U. Schäfer et al. 2017; C. P. Abod et al. 2019; R. Li et al. 2019). Despite this, these studies consistently show similar mass distributions, which can be approximated by a steep exponential power law at the high-mass end and a gradual taper in the small-mass range.

This section presents the cumulative distribution of the absolute magnitude[18] H for our sample of CC objects. Recent works demonstrate (J. Kavelaars et al. 2021; K. J. Napier et al. 2024) that the differential SD of CCs is consistent with an exponentially tapered power-law form—a feature of numerical simulations of planetesimal formation via SI + PCC. The depth of these surveys, however, was limited to $H_r \sim 8.3$ for OSSOS and $H_r \sim 10.5$ for DEEP. G. M. Bernstein et al. (2004) provided an anchor on the faint-end slope. This JWST survey is the first to add objects with $H_r > \sim 10.5$, and pushes the faint limit to nearly $H_r = 14$. While the OSSOS and DEEP results agree well at brighter magnitudes ($H \lesssim 11$), increasing uncertainties arise at fainter magnitudes due to limited small-object statistics. Our survey extends beyond these previous limits, and in Figure 20, we superpose our cumulative H distribution on the extrapolated best-fit exponentially tapered power laws from J. Kavelaars et al. (2021) and K. J. Napier et al. (2024). We find that the cumulative distribution of our detected CCs is shallower than those reported in the two studies, suggesting that the previously proposed exponentially tapered power-law model may not fully represent the SD of the smallest objects in this population.

[18] $H = m - 5\log_{10}(r(r-1)) + 2.5\log_{10}\phi(\alpha)$, where r is the heliocentric distance in astronomical units and $\phi(\alpha)$ is the phase function. At quadrature, the geocentric and heliocentric distances are similar, so the term $r(r-1)$ becomes r^2 and $\alpha = 0°$, so $\phi = 1$ (E. Bowell et al. 1989; D. W. Hughes 2003).

9. Mass of the Cold Kuiper Belt

Given the surface density $\Sigma(m)$, we are able to estimate the total mass of the CC belt as

$$M_{\rm tot} = M_{23}\frac{\Omega}{f}\int_{m=14}^{m=35}\Sigma(m)\,10^{-0.6(m-23)}dr \times \left\langle \left(\frac{p}{p_{\rm nom}}\right)^{-3/2}\left(\frac{d}{d_{\rm nom}}\right)^{6}\left(\frac{\rho}{\rho_{\rm nom}}\right)\right\rangle, \quad (23)$$

where Ω is the surveyed solid angle, and f is the fraction of the TNO population located within this surveyed area. Using the OSSOS++ model, which is the leading dynamical TNO model accounting for both the orbital structure and SD (J.-M. Petit et al. 2011, 2017; M. Alexandersen et al. 2016; M. T. Bannister et al. 2018), we estimate the fraction of CCs per square degree at the epoch and location of our observation to be $f \approx 3.8 \times 10^{-4}$. The normalization constant $M_{23} = 7.8 \times 10^{18}$ kg corresponds to the mass of a TNO with apparent magnitude $m_r = 23$, assuming canonical values for albedo, density, and distance.

The angle brackets denote the mean value of all TNOs at a given magnitude. For the CC population, the average heliocentric distance is $d \sim 43.8$ au (M. T. Bannister et al. 2018; K. J. Napier et al. 2024), while the average albedo is also

Table 6
Best-fit Differential Surface Density Models

Sample	Single Power Law		Rolling Power Law				Exponentially Tapered Power Law				
	α	m_0	α_1	α_2	Σ_{23}	ΔBIC	α	β	m_0	m_B	ΔBIC
Hot A	$0.24^{+0.11}_{-0.09}$	$20.89^{+2.16}_{-3.64}$	$0.90^{+0.09}_{-0.09}$	$-0.10^{+0.02}_{-0.02}$	$0.40^{+0.10}_{-0.10}$	0.2	$0.18^{+0.07}_{-0.06}$	$0.20^{+0.23}_{-0.16}$	$16.39^{+3.67}_{-6.60}$	$24.89^{+3.35}_{-6.18}$	3.1
Cold A	$0.27^{+0.09}_{-0.08}$	$20.96^{+1.76}_{-2.78}$	$0.97^{+0.09}_{-0.08}$	$-0.09^{+0.02}_{-0.02}$	$0.20^{+0.09}_{-0.08}$	2.1	$0.21^{+0.08}_{-0.06}$	$0.21^{+0.26}_{17}$	$17.03^{+3.47}_{-5.47}$	$24.44^{+4.23}_{-6.02}$	4.9
Full A	$0.28^{+0.07}_{-0.06}$	$20.33^{+1.47}_{-1.95}$	$0.92^{+0.08}_{-0.08}$	$-0.08^{+0.02}_{-0.02}$	$0.41^{+0.09}_{-0.09}$	0.6	$0.22^{+0.07}_{-0.06}$	$0.23^{+0.21}_{-0.17}$	$16.72^{+3.12}_{-5.33}$	$25.08^{+2.51}_{-5.37}$	4.3

Note. ΔBIC is similar to that described in Table 3.

found to be $p \sim 0.15$ (W. C. Fraser et al. 2014; P. Lacerda et al. 2014). The material density of an object is a function of its size, with smaller TNOs often having densities as low as 500 kg m^{-3} and larger objects being as dense as 3000 kg m^{-3} (W. Grundy et al. 2015). For simplicity, we assume a density corresponding to a sphere of pure water ice, $\rho \sim 900$ kg m^{-3}.

Using the best-fit parameters for the Cold AB sample and assuming a rolling power-law SD, we find that the total mass of the CC belt is 0.01 $M_\oplus \times \left(\frac{p}{0.15}\right)^{-3/2} \times \left(\frac{d}{43.8}\right)^{6} \times \left(\frac{\rho}{900\ \mathrm{kg\ m^{-3}}}\right)$. Note that we only calculate for the total mass of the CC population, as the total mass of the HTNOs is more uncertain due to poor constraints on their vertical and radial distribution, although G. M. Bernstein et al. (2004) suggested that the mass of objects within 50 au is comparable to that of the CKB. We also choose to report the mass of the rolling power law to provide direct comparison with other recent works (G. M. Bernstein et al. 2004; K. J. Napier et al. 2024). With our assumptions, the integrated mass faintward of our detection limit is $\sim 7 \times 10^{-5}\, M_\oplus$, much smaller than the mass contained in larger objects. This implies that the CC belt mass uncertainty is no longer driven by uncertainty in the LF, but in the albedo and density assumptions (W. C. Fraser et al. 2014).

10. Discussion and Conclusions

In this paper, we have presented detections from the first TNO-dedicated survey using JWST. By using the shift-and-stack technique, we are able to achieve a limiting magnitude (40% efficiency) of $m_{\mathrm{F150W2}} = 28.6$ for sample A and $m_{\mathrm{F150W2}} = 28.8$ for sample A+B (assuming F150W2 $- r \approx 1.0$, this translates to $m_r \approx 29.6$ mag and $m_r \approx 29.8$ mag, respectively) over a sky area of 0.05 deg^2. This is *by far* the deepest TNO survey to date, with roughly a visible magnitude deeper than the landmark HST survey by G. M. Bernstein et al. (2004), which had a limiting magnitude of $m_r \sim 28.9$ and covered 2.5 × more survey area. Our data yield 23 three-epoch detections and 7 two-epoch detections, with magnitudes $24.1 < m_{\mathrm{F150W2}} < 29.3$ and heliocentric distances 28.3 au $< r \lesssim$ 133.6 au—all new discoveries except 2015 GK56, which was purposely targeted. In addition, we identify 13 single-epoch detections that aren't static transients. None of these new discoveries exhibit lightcurve indicative of binary TNOs, in agreement with expectations scaled from the SSOLS dataset. Using these discoveries, we determine the faint-end magnitude distributions for both the cold and hot TNO populations, as well as for the Kuiper Belt as a whole.

The cold population of our nominal sample (sample A+B) exhibits a shallow slope of $\alpha = 0.29^{+0.07}_{-0.08}$, comparable to that of the hot component, $\alpha = 0.28^{+0.09}_{-0.11}$. The observed LFs suggest that both populations have similar SDs. The slope derived for the full TNO sample is also consistent with those of the individual components. These consistently shallow slopes across all populations indicate that the majority of the Kuiper Belt's mass is concentrated in objects near the break diameter, ∼200 km.

Multiple lines of evidence suggest that the cold and hot TNO populations are genetically distinct, having likely formed in different regions of the planet-forming disk and under different physical conditions. For instance, the CCs are predominantly very red in color, while HTNOs display a wide range of colors—a contrast that is generally interpreted as indicative of primordial compositional differences between the distinct regions in which they formed (M. E. Brown et al. 2011; W. C. Fraser & M. E. Brown 2012; M. E. Schwamb et al. 2019; D. Nesvorný et al. 2020; L. E. Buchanan et al. 2022; W. C. Fraser et al. 2023; P. H. Bernardinelli et al. 2025; M. N. De Prá et al. 2025; E. Hénault et al. 2025; N. Pinilla-Alonso et al. 2025). CCs are thought to have formed near their current locations, while HTNOs likely originated closer to the Sun before being scattered outward by the migration of the giant planets (A. Morbidelli et al. 2005; K. Tsiganis et al. 2005; H. F. Levison et al. 2008; D. Nesvorný & A. Morbidelli 2012; D. Nesvorný 2015). The region where CCs reside is also thought to have been relatively less dense, causing Neptune to halt its outward migration at 30 au (R. S. Gomes et al. 2004; D. Nesvorný 2018), and to have undergone relatively modest collisional evolution, thereby preserving surface volatiles responsible for their red coloration (A. Y. Abedin et al. 2022). The prevalence of binaries in the CC belt further supports the idea that CCs are formed *in situ* and are therefore primordial, as substantial dynamical excitation during the giant planet migration would likely have disrupted such systems (A. H. Parker & J. J. Kavelaars 2010; K. S. Noll et al. 2020). In contrast, the higher density of the inner disk from which HTNOs originated would have exposed them to more intense collisional grinding, further altering their surface properties (D. Nesvorný 2018). Despite these strikingly different formation environments and evolution histories, our results indicate that CCs and HTNOs nevertheless share a remarkably similar SD.

The similarity of slopes across the two populations we observe in this work ($H_r \simeq$ 8.3–13.4 mag) is consistent with J.-M. Petit et al. (2023), who reported comparable SD shapes for both populations in the absolute magnitude range $H_r \simeq$ 5.5–8.3 mag. The very similar SDs from $H_r \sim 5$ down to the observable limit suggest that both populations likely underwent analogous formation mechanisms, resulting in nearly identical SDs despite those processes occurring in very different conditions. That is to say, the underlying process responsible for planetesimal formation (potentially SI+PCC) appears to be largely insensitive to local disk conditions (e.g., temperature and density gradients), resulting in a consistent SD regardless of environments. Consequently, the indistinguishable shallow slopes imply that the hot population, much like the cold one, seems to have retained its primordial SD down to diameters of ∼10 km.

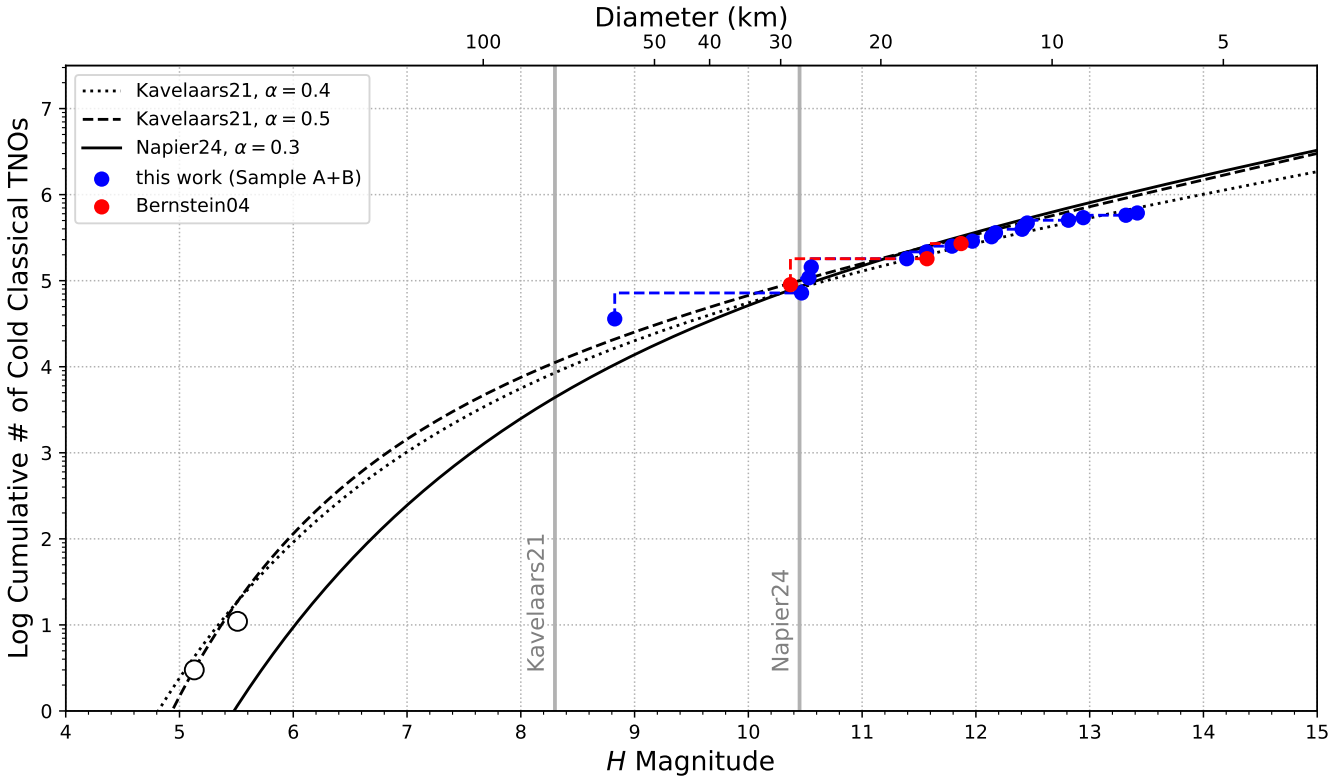


Figure 20. The H_r cumulative distribution of the cold main Kuiper Belt. Blue circles show the scaled cumulative distribution of the CCs from sample A+B of our survey, while red circles represent the cumulative distribution of CCs detected by G. M. Bernstein et al. (2004). The dotted and dashed lines indicate the best-fit power-law models from J. Kavelaars et al. (2021), corresponding to slopes of $\alpha = 0.4$ and $\alpha = 0.5$, respectively. The solid black line shows the best-fit exponentially tapered function from K. J. Napier et al. (2024), scaled to match the full ecliptic population. White open circles mark the cumulative counts of 3 ($Hr \sim 5.13$) and 11 ($Hr \sim 5.51$) main-belt cold objects, as reported in the MPC database. The vertical lines indicate the observing limit from each study.

The shallow slopes observed in both populations are consistent with the slope ($\alpha \sim 0.4 \pm 0.04$) of impactors above the turnover in the SD inferred from the Pluto and Charon cratering record (K. N. Singer et al. 2019). This behavior hints at the idea that the SI+PCC mechanism requires a minimum mass or size threshold for gravitational collapse to occur efficiently. That turnover may mark the point at which local pebble concentrations become dense enough to overcome turbulent diffusion, enabling the formation of bound planetesimals. Below this critical size, the collapse mechanism could become much less effective, leading to a relative deficit of smaller bodies.

Moreover, the fact that we find the HTNOs slope to be shallow, despite originating from a denser and collision-heavy environment, places further constraint on the nature and timescale of its evolutionary process. In particular, this suggests that dynamical excitation and subsequent collisional evolution were not prolonged or intense enough to significantly alter the original distribution of bodies larger than 10 km.

In the absence of direct size measurements from stellar occultations, the geometric albedos (ρ) of smaller TNOs are inferred from the median albedos of their dynamical classes, which have been derived from observations of larger objects (J. Stansberry et al. 2007; P. Lacerda et al. 2014; E. Vilenius et al. 2014). This then introduces a bias in size estimates, particularly for the CC population, making them appear bigger[19] than they may actually be. The result that fainter CCs and HTNOs exhibit similar slopes may indicate that their albedos converge at smaller sizes, implying more comparable reflectivities in that regime.

If the hot and cold populations exhibit different albedo distributions across the observable size range, then the similarity of their magnitude distributions implies a similar shape of the underlying SDs, but they are offset by an amount comparable to the magnitude of reflectance difference driven by their albedos. It has been suggested that on average, smaller CC TNOs are redder than larger bodies (S. D. Benecchi et al. 2011). If this suggestion is true, it is conceivable that albedos may also have a size dependency such that the difference in average albedos of the hot and cold populations vanishes at smaller sizes. It is likely that evidence for such a trend will arise with further compositional and thermal observations with a focus on smaller objects.

Although the hot and cold TNO populations currently appear to share the same SD across their entire observable range, this inference is drawn from a sample of only about 1000 objects. Characterized TNO discoveries expected from the upcoming Legacy Survey of Space and Time (LSST; Ž. Ivezić et al. 2019) will increase this sample by more than an order of magnitude (e.g., J. A. Kurlander et al. 2025), enabling a detailed search for any differences between the SDs of the hot and cold populations. In particular, it will be possible to search for any small variations in the SD due to the enhanced collisional grinding expected of the hot population since its formation. However, despite its unprecedented sample size, LSST's limiting magnitude is $H_r \sim 8$ mag and will not constrain the faint end of the TNO SD as in this work.

[19] The absolute magnitude (H) scales as $H \propto -2.5\log(\rho D^2)$, where D is the object's diameter.

Finally, we note the ongoing efforts to assess the population of objects beyond the main Kuiper Belt. Recent work has suggested that there may be an increase in density of objects beyond ∼80 au as compared to objects at closer distances (W. C. Fraser et al. 2024). Scaling those results to our survey, and assuming an LF similar to that observed here, we would expect roughly 1 detection at ∼80 au; and we have detected one object at these distances. We point out, however, that our current sample of distant such bodies remains much too small to provide meaningful constraints.

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Appendix A
ResNet Configurations and Acceptance Threshold Selection

We explored 10 different ResNet model configurations and ultimately settled with the model that achieved the faintest detection depth and highest peak detection efficiency. Comparison of the performance of these 10 models is shown left of Figure 21. The model with the optimum performance is Model 10. We then investigate the best acceptance threshold for Model 10 by evaluating how the number of vetted predictions varies with the threshold (see right of Figure 21), identifying the point where a lower threshold introduces excessive false positives while a higher threshold becomes overly stringent.

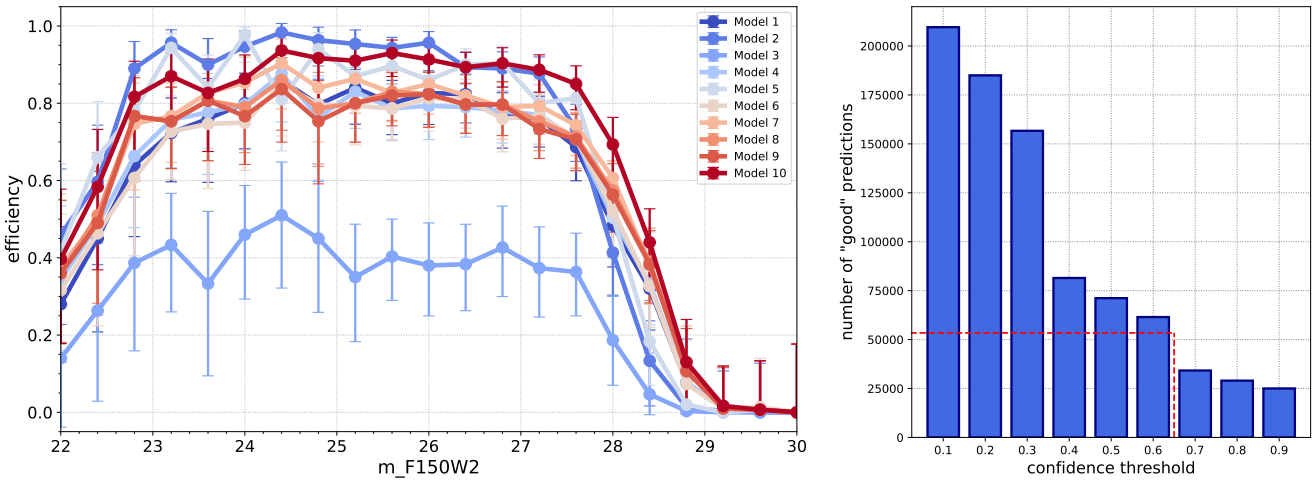


Figure 21. Left: detection efficiency of 10 ResNet models. We find Model 10 to be the most ideal. Right: number of objects vetted as "good" source by Model 10 as a function of acceptance threshold. We eventually settled with 0.65.

Appendix B
Information on Individual TNOs

We describe here the steps that were used to discover and determine orbital elements for each of the two- and three-epoch TNOs listed in Table 1. Those that are not listed here were discovered through "normal procedures," meaning that three epochs' tracks were found, linked, and fit to an orbit using the procedures described in Sections 4 and 5.1.

JPB01. The S/N = 51 apparition in epoch 2 was misclassified as spurious by the ResNet, probably because of overtraining on implants with a different PSF. It is considered part of sample A.

JPB02. The S/N = 7.8 apparition in epoch 1 was classified as spurious by ResNet, and linked in the search that bypassed ResNet. Since the implants used to determine detection efficiency and ensure against false positives were filtered by ResNet, we cannot ensure the reality of this source nor use it as part of sample A for population statistics. Since the epoch 3 detection has S/N < 8, it is also excluded from sample B. We retain it in the auxiliary files as an illustration of an uncertain detection.

JPB03. The S/N = 5.9 apparition in epoch 3 was classified as spurious by ResNet, and linked in the search that bypassed ResNet. This is even fainter than JPB02 and is excluded from both samples A and B for the same reasons.

JPB04. The duplicate detection at S/N = 178 in epoch 1 was misclassified by ResNet. The object remains in sample A since it was discovered in three epochs via normal procedures.

JPB05. The S/N = 40 apparition in epoch 3 was misclassified as spurious by the ResNet. Like JPB01, this is probably a ResNet blunder, and the object is considered part of sample A.

JPB23. This is 2015 GK_{56}, a previously known TNO that was placed in the FOV intentionally. It is therefore not a random sampling of the sky population and is omitted from the population-statistics samples.

JPB24. The epoch 2 apparition at S/N = 7.4 was not found by KBMOD because of residuals from the subtraction of the bright galaxy in its background. This source is *not* included in sample A, because the calibrating implanted sources were subject to the same sort of failure. It does, however, enter sample B, since its other two epochs were found at S/N > 8. The distance is well determined from the recovery of epoch 2 using SMP.

JPB25. This object has pulled a "vanishing act." It was found once in epoch 2 and twice in epoch 3 by normal procedures at S/N = 8.6, 12, and 8.7. Linking these two epochs leaves a very wide range, $55 < d_{\rm bary} < 280$ au, as possible (meaning that this is not a cold source). At any such distance, the object was imaged by an unobscured region of the NIRCAM detector B2 in epoch 1, but all attempts to recover the source failed, requiring this apparition to have S/N < 4.5. We conclude that either this is a TNO that has flux variations of a factor $\gtrsim$2, or the linkage is spurious. We choose to omit this TNO from sample A (since it was not found in three epochs), but include it in sample B (since the other two epochs have S/N > 8), despite some doubt that it is real. The distance is, in any case, too large to be a CC.

JPB26. This source was found twice in epoch 1 and once in epoch 2 at 11.3 < S/N < 15.2, but leaves no trace in epoch 3. It could avoid detection by having $40.3 < d_{\rm bary} < 48.6$ au, in which case it would have been in a gap between detectors. Otherwise it would have appeared in unobscured regions of a NIRCAM SW detector, where it would have needed to be $\gtrsim 3\times$ fainter than in the other epochs to avoid detection, which we consider unlikely. Under the former interpretation, this object belongs to sample B. This distance range and its inclination $i < 2^\circ$ place it nominally among the cold sources, albeit with $e_{\rm min} > 0.25$.

JPB27. The object is missing from epoch 2 despite S/N ≈ 90 detections in epochs 1 and 3. The only viable explanation is that the object has $42.4 < d_{\rm bary} < 42.6$ au, which causes it to fall into a gap between two SW detectors. Indeed, the object is detectable in HST WFC3 exposures (A. Morgan et al. 2026), which allow a precision determination of $d_{\rm bary} = 42.49$, placing this source into sample B and also among the dynamically cold bodies.

JPB28. Detections at S/N ≈ 15 in epochs 1 and 2 indicate that the object would have been in a detector gap in epoch 3 if $43.2 < d_{\rm bary} < 48.0$ au. We take the absence of detection to exclude other distances. This places the source in sample B, with cold dynamics.

JPB29. Detections at S/N ≈ 19 in epochs 1 and 3 indicate that this TNO was outside the NIRCAM fields of view in epoch 2 for any viable distance. Observed motion rates of motion imply $41.2 < d_{\rm bary} < 46.9$ au, placing this source in sample B, with cold dynamics.

JPB30. Detections at S/N ≈ 14 in epochs 2 and 3 indicate that this TNO was outside the NIRCAM fields of view in epoch 1 for any viable distance. Observed rates of motion constrain the distance poorly, $42.8 < d_{\rm bary} < 60.2$ au, although the inclination is always too high to be a CC. The object is in sample B.

Appendix C
Available Data

The NIRCAM images collected by the program are available from the STScI MAST archive under program GO-1568, which can be accessed via doi:10.17909/ekq3-rz42. We also make available at doi:10.11570/26.0008 additional files that facilitate further analyses of the moving/transient sources that we have discovered in these images. A full inventory of the auxiliary files and the meaning of all table columns is in the `readme.md` file is included in the repository. Briefly, the auxiliary files include:

1. Tables including cutouts from the individual integrations' images around all of the detected objects in Table 1, plus the positions, rates, and fluxes, and pixelized models that best fit these images in each epoch. Both SW and LW data are included.
2. A library of figures giving the key statistics and showing the cutouts and stacked cutouts for each track.
3. A table giving the results of orbit-level SMP fitting (Section 5) to each TNO appearing in three epochs. This includes the individual integrations' cutouts with the models jointly fit to all epochs; the best-fit orbital parameters and fluxes, plus their uncertainties; the epoch-averaged and fully averaged fluxes/magnitudes and LW/SW flux ratios; and their classifications into our samples A and B, and dynamically cold samples.
4. A library of figures for each three-epoch TNO detection in the format of Figure 12.

5. A table giving the results of the limiting orbit fitting to the two-epoch TNOs, plus summarization of their SW and LW fluxes in their observed epochs.
6. A table summarizing the fluxes and rates of motion of the single-epoch detections listed in Table 2.
7. Two full mosaic images of all 20 tiles in the LW and SW filters.
8. A folder containing background- and static-source-subtracted images for each integration, epoch, tile, and dither.

Appendix D
Derivation of Likelihood Function

For an assumed sky surface density of TNOs versus mean magnitude

$$\Sigma(m,\,\theta) = \frac{dN}{dm\,d\Omega} \tag{D1}$$

with free parameters θ, we wish to express the Bayesian posterior probability $p(\theta|\{\hat{m}_i\}$ of the parameters conditioned on the survey's detection of TNOs with measured magnitudes m_i, $i = 1, \ldots, M$. Expressions for this have been derived—with different results—by G. M. Bernstein et al. (2004) and T. J. Loredo (2004), among others. Here we derive what we hope are clarifying results.

Under Bayes' theorem, we will have

$$p(\theta|\{\hat{m}_i\}) \propto \mathcal{L}(\{\hat{m}_i\}|\theta)p(\theta), \tag{D2}$$

with the last term being a prior on the parameters of Σ. We concentrate on deriving the likelihood term $\mathcal{L}$.

In our survey and many others, there is a detection/selection step followed by a measurement step. For any TNO within the survey area with true mean magnitude m, we can denote successful detection/selection by the condition s. One outcome, $\tilde{s}$, is that the object is not found at all, or is found and fails the selection criteria. We denote $\eta(m) = p(s|m)$ the probability of successful selection. For every *selected* source, we also measure a magnitude $\hat{m}$ for the source, so there is a joint probability

$$p(\hat{m},\,s|m) = p(\hat{m}|s,\,m)\eta(m). \tag{D3}$$

T. J. Loredo (2004) assumes that selection is made on the basis of the $\hat{m}$ value alone, so that for any value $\hat{m}$ in some region S, we can write $p(s|\hat{m}) = 1$, and we can create an unambiguous function $p(\hat{m}|m)$ for any $\hat{m} \in S$. But in many surveys, there are stochastic characteristics of astronomical observations that influence selection without changing $\hat{m}$, such as whether the orbit brings the source into functioning areas of the detectors, or atop bright sources, and how image noise fluctuations orthogonal to $\hat{m}$ affect our ResNet outputs. In general, there is no universal expression as $p(\hat{m}|m)$ since at least *some* TNOs at m will never get a measured magnitude.

Assuming each TNO's selection and measurement to be independent of other TNO's existence or outcome, and a surveyed solid angle Ω with uniform expected density, we obtain the standard Poisson formulation of the likelihood:

$$\mathcal{L}(\{\hat{m}_i,\,s_i\}|\theta) = e^{-\bar{N}(\theta)} \prod_i\, \Omega \int dm_i\, p(\hat{m}_i,\,s_i|m_i)\Sigma(m_i,\,\theta) \tag{D4}$$

$$\propto e^{-\bar{N}(\theta)} \prod_i\, \Omega \int dm_i\, p(\hat{m}_i|s_i,\,m_i)\eta(m_i)\Sigma(m_i,\,\theta), \tag{D5}$$

$$\bar{N}(\theta) \equiv \int d\hat{m} \int dm\, p(\hat{m},\,s|m)\Omega\Sigma(m,\,\theta) \tag{D6}$$

$$= \Omega \int dm\, \Sigma(m,\,\theta) \int d\hat{m}\, p(\hat{m},\,s|m) \tag{D7}$$

$$= \Omega \int dm\, \Sigma(m,\,\theta)\eta(m). \tag{D8}$$

We calculated the selection rate $\eta(m)$ in Section 4.3.3 using implanted sources, as is typical for most analyses, and enables calculation of $\bar{N}(\theta)$ for any choice of θ using the last equation.

Knowledge of $\eta(m)$ is, however, formally insufficient to calculate $\mathcal{L}$. There are two possible approaches. One is to use implanted sources to characterize the joint distribution $p(\hat{m},\,s|m)$, and proceed using Equation (D4). This, however, requires a larger number of simulated sources than estimating the univariate function $\eta(m)$, so we have not done this.

A second approach is to know or approximate the conditional distribution $p(\hat{m}|s,\,m)$ and use Equation (D5). Injected sources could be used for this, with the same caveat that the joint distribution requires more injections. We choose instead to make the approximation that $p(\hat{m}|s,\,m)$ is narrow, i.e., the measurement noise on magnitude is small compared to the scale of variation of Σ against which it is being integrated. This yields our adopted form of the likelihood,

$$\mathcal{L}(\{\hat{m}_i,\,s_i\}|\theta) \propto e^{-\bar{N}(\theta)} \prod_i\, \eta(m_i)\Sigma(m_i,\,\theta). \tag{D9}$$

D.1. Comparison with Loredo (2004)

We note that neither of our exact forms in (D4), (D5), nor our approximate form in (D9), match the results of T. J. Loredo (2004; hereafter, L04). L04 derives their Equation (8)

$$\mathcal{L}(\{\hat{m}_i\}|\theta) = e^{-\bar{N}(\theta)} \prod_i\, dm\, p(\hat{m}_i|m)\Sigma(m,\,\theta). \tag{D10}$$

This form lacks any specific conditioning on s because it is assumed that selection is a deterministic Heaviside function of $\hat{m}$, and every TNO has some measured value $\hat{m}$ with known $p(\hat{m}|m)$, such that:

$$p(\hat{m},\,s|m) = \begin{cases} p(\hat{m}|m) & \hat{m} < m_{\rm lim} \\ 0 & \hat{m} > m_{\rm lim}. \end{cases} \tag{D11}$$

With this substitution, our Equation (D4) reduces to L04's form.

As noted earlier, there is the possibility that the selection is not binary over the space of $\hat{m}$; $p(\hat{m}|m)$ is not always well defined without being joint with or conditioned on selection s. But if it is well defined, it is not the appropriate quantity for the likelihood integration. As an example, consider the absence of measurement errors so that $p(\hat{m}|m) = \delta(\hat{m} - m)$. In this case, the distribution of observed magnitudes will be $p(\hat{m}) = \eta(m)\Sigma(m)$, and the likelihood clearly takes the Poisson form

$$\mathcal{L}(\{\hat{m}_i\}|\theta) = e^{-\bar{N}(\theta)} \prod_i\, \Omega\eta(\hat{m}_i)\Sigma(\hat{m}_i,\,\theta), \tag{D12}$$

containing a factor of $\eta(m)$ in the integrand that is missing from L04's form, but is present in Equation (D5). The missing

factor can be traced to a small error in L04 , where, when generalizing their Equation (5), a factor of $\eta(m)$ is added for the mean $\bar{N}$ but is not also added under the product part of the Poisson likelihood in his Equation (8).

To summarize, the correct form of the observational likelihood (and hence the Bayesian posterior derived from it) is given by either Equation (D4) using the joint $p(\hat{m}, s|m)$ or Equation (D5) using the equivalent quantity $p(\hat{m}|s, m)\eta(m)$. If the joint or conditional distribution is not available, L04's form, Equation (D10), is equivalent in the limit where the selection probability is a binary function of the observed $\hat{m}$, i.e., the measurement noise on m determines the selection. A different limit, when the measurement noise is small compared to the range of m over which selection $\eta(m)$ and $\Sigma(m)$ vary, admits the approximation yielding Equation (D9), used by G. M. Bernstein et al. (2004). We point out that in the current work, these subtleties in the treatment of the joint distribution make an insignificant difference to the inference, since there are few counts in the magnitude regime where $\eta(m)$ transitions from 0 to its maximum value.

Appendix E
Object Classification and Spectral Types

Table 7 lists the objects' orbital and spectral classification. We limit our compositional estimates to objects with three epoch detections, since their orbital solutions are well constrained.

Table 7
Object Classification and Spectral Types

Object	CC?	Spectral Type
JPB01	Yes	Organics
JPB04	Yes	Organics
JPB05	Yes	Organics
JPB06	Yes	Organics
JPB07	Yes	Organics
JPB08	Yes	Organics
JPB09	No	H_2O/CO_2
JPB10	No	H_2O
JPB11	No	H_2O/CO_2
JPB12	Yes	Organics
JPB13	Yes	Organics
JPB14	No	CO_2
JPB15	Yes	Organics
JPB16	Yes	Organics
JPB17	No	Organics
JPB18	No	H_2O
JPB19	No	H_2O
JPB20	Yes	Organics
JPB21	Yes	Organics
JPB22	No	H_2O
JPB23	Yes	Organics
JPB27[a]	Yes	Organics

Notes. Objects labeled as H_2O/CO_2 have uncertainties large enough that they could either be H2O or CO_2 type. Therefore, we use the mean $r - F150W2$ value of the two types for those objects.

[a] This object was detected in two JWST epochs and independently in HST WFC3 images (see Section 5.3), allowing us to constrain its orbital solution; nevertheless, we exclude it from our LF fits.

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